GSC FIELD APP GUIDE
Version 3.1

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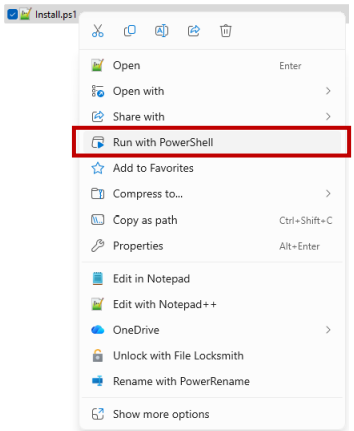
INSTALLING ON WINDOWS / INSTALLATION SUR WINDOWS

1. Download the latest Windows version from https://github.com/NRCan/GSC-Field-Application/releases.
2. Expand the Assets section near the bottom of the page

3. Download the Windows version to your PC, tablet or phone device.
4. Extract the zip file contents and navigate to that location.
5. Run GSCFieldApp_3.1.0.0_x64.msix (or equivalent) to install the GSC Field App. Alternatively, you can choose to run Install.ps1 with Powershell.



INSTALLING ON ANDROID / INSTALLATION SUR ANDROID

1. Download the latest Android version from https://github.com/NRCan/GSC-Field-Application/releases.
2. Expand the Assets section near the bottom of the page.

3. Copy the .apk file to your Android device (probably using a USB cable or drive)
4. On your Android device, run the .apk file to install the GSC Field App

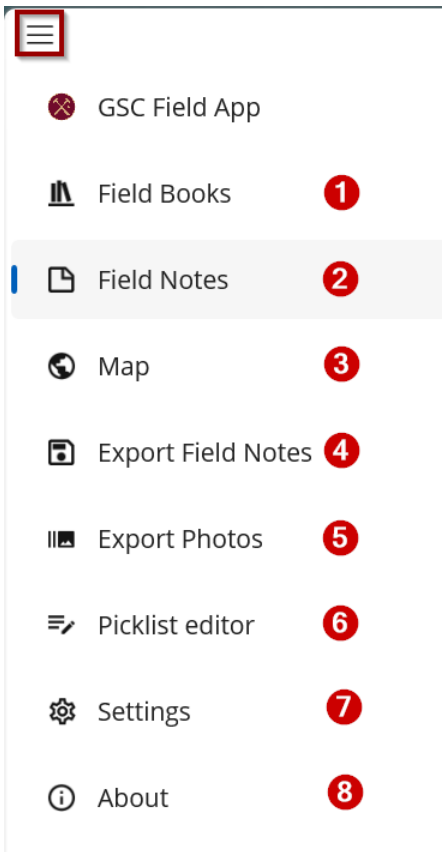
NOTE: Some Samsung phones experience the Ruler button not being fully visible. Correct the issue with this fix:

1. Open Settings.
2. Navigate to Display > Screen Zoom.
3. Set the slider to the lowest value possible.

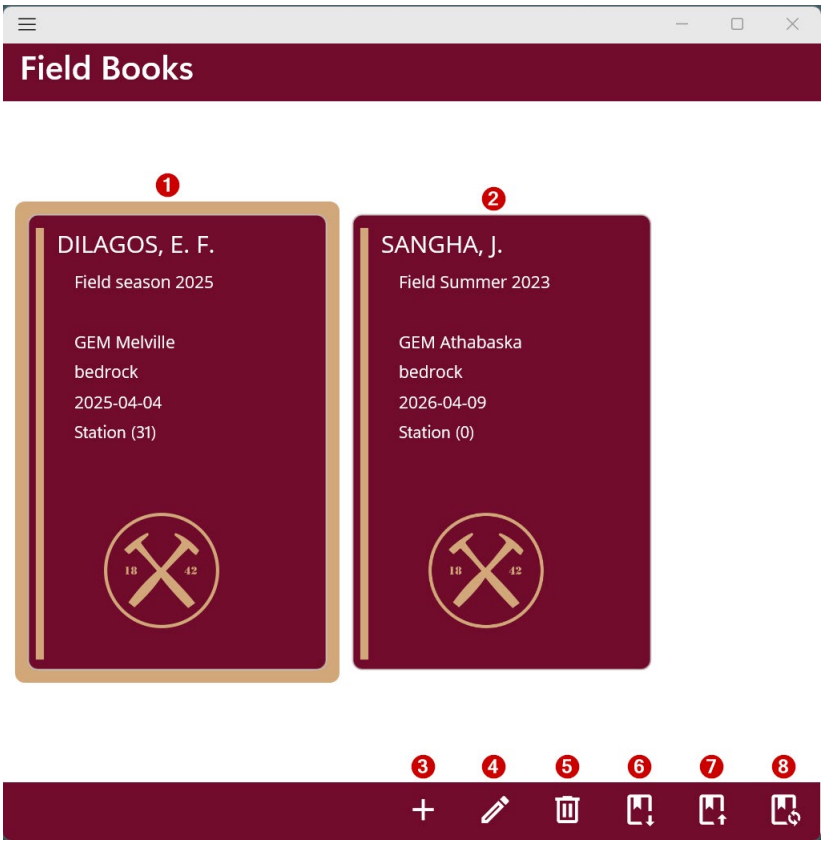
MAIN MENU / MENU PRINCIPAL

Field Books – a complete field book for a specific area of interest. This section contains all of the organizational metadata about a project.

1. **Field Books** – Use Field Books to organize separate projects.
2. **Field Notes** – the detailed area where individual stored records on Stations, Photographs, Samples, etc. can be examined and revised.
3. **Map** – the map of the research area with all of the data plotted.
4. **Export Field Notes** – allows the user to create a full backup of the current field book.
5. **Export Photos** – allows the user to create a backup of the photos from the current field book.
6. **Picklist Editor** – allows the user to modify the picklists entries, addition/subtraction/sorting the display order of certain key terms for an individual project.
7. **Settings** – turn off/on certain tables as needed.
8. **About** – Contact information for this app.



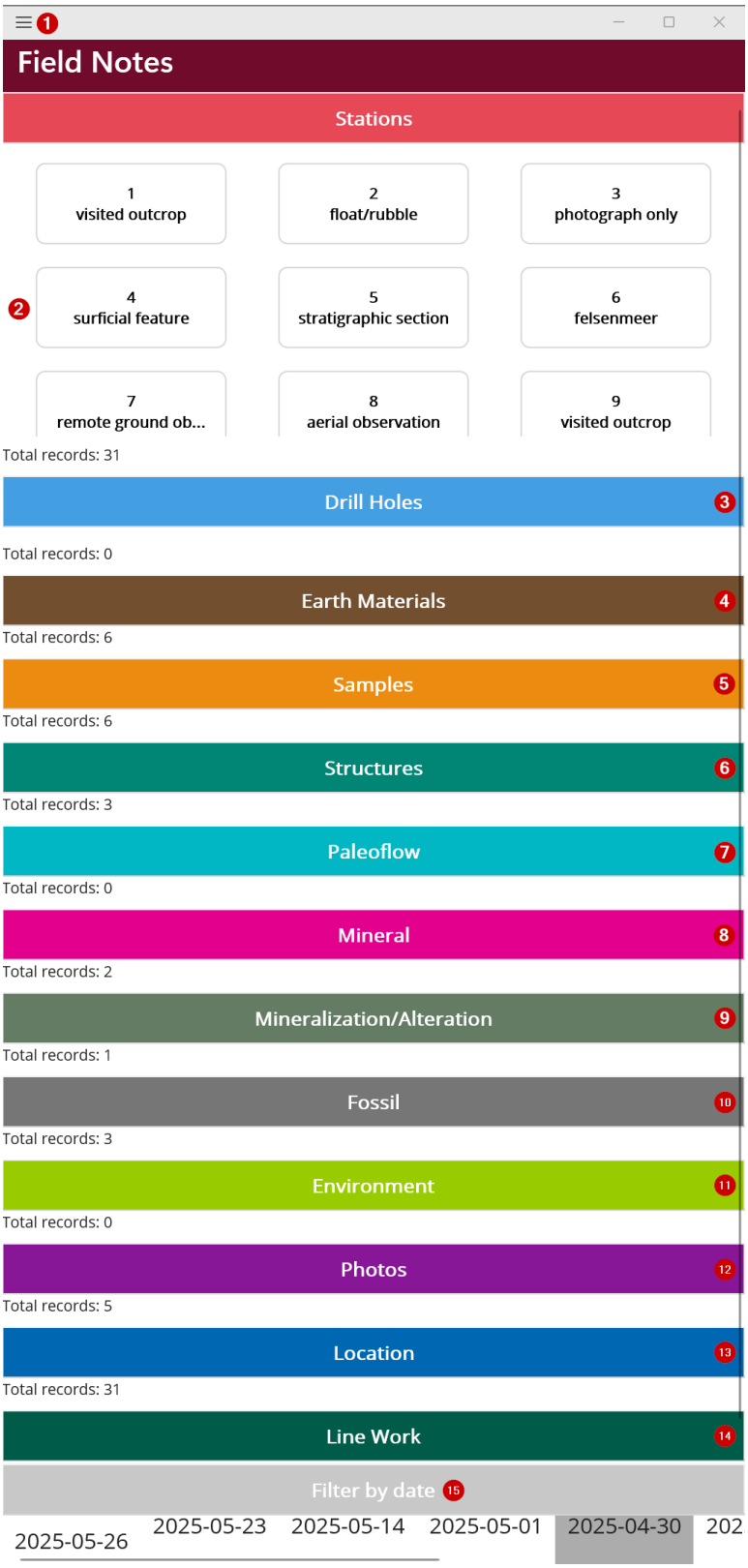
FIELD BOOKS / LIVRES DE TERRAIN



Navigation Bar → Summary of application pages.

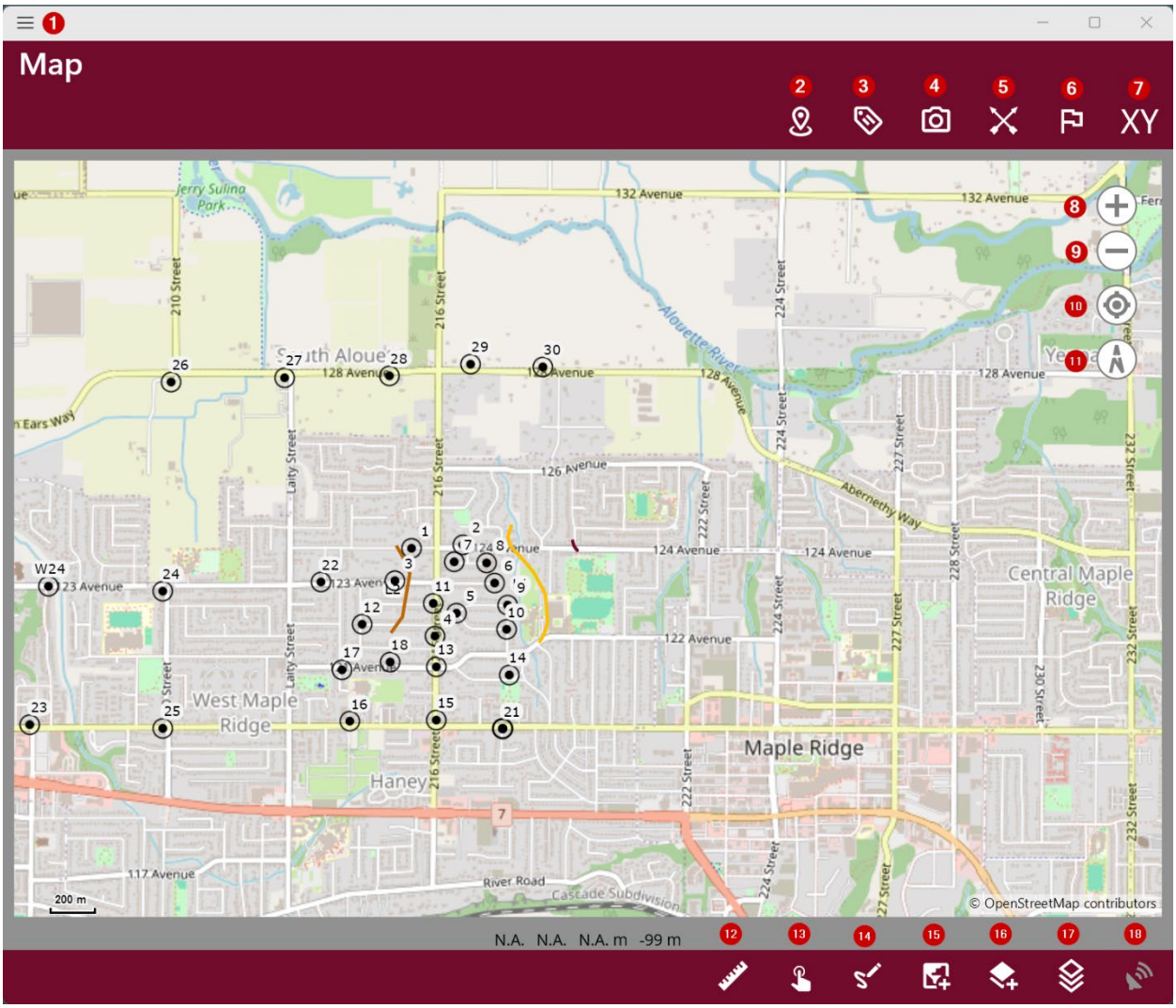
1. **The currently selected field book.**
2. **Another field book.**
3. **Add a new field book:** add a new field book.
4. **Edit selected field book**: allows you to modify the metadata for the current field book.
5. **Delete selected field book**: deletes the current field book. Do not do this unless you already have a backup as there is no way to undo a deletion of a field book.
6. **Download selected field book**: Creates a backup of the current field book.
7. **Upload a field book**: Restores a backup into the current field book (.gpkg or .zip)
8. **Update selected field book to current application version**: updates the selected field book to any changes in the database structure (requires a new XML structure file).

FIELD NOTES / NOTES DE TERRAIN



1. Main menu (Field Books, Field Notes, the Map View, Export Field Notes, Export Photos, Picklist Editor, Settings, About).
2. A list of station records in the Field Notes view.
3. Expand Drill Holes to see records (none in this case).
4. Expand Earth Materials to see records (1).
5. Expand Samples to see records (1).
6. Expand Structures to see records (0).
7. Expand Paleoflow to see records (0).
8. Expand Mineral to see records (1).
9. Expand Mineralization/Alteration to see records (1).
10. Expand Fossil to see records (1).
11. Expand Environment to see records (3).
12. Expand Photos to see records (1).
13. Expand Location to see records (9).
14. Expand Line Work to see records (0).
15. Click on the Traverse Date tabs to look at traverse data collected on different days.

MAP PAGE / APPLICATION NAVIGATION

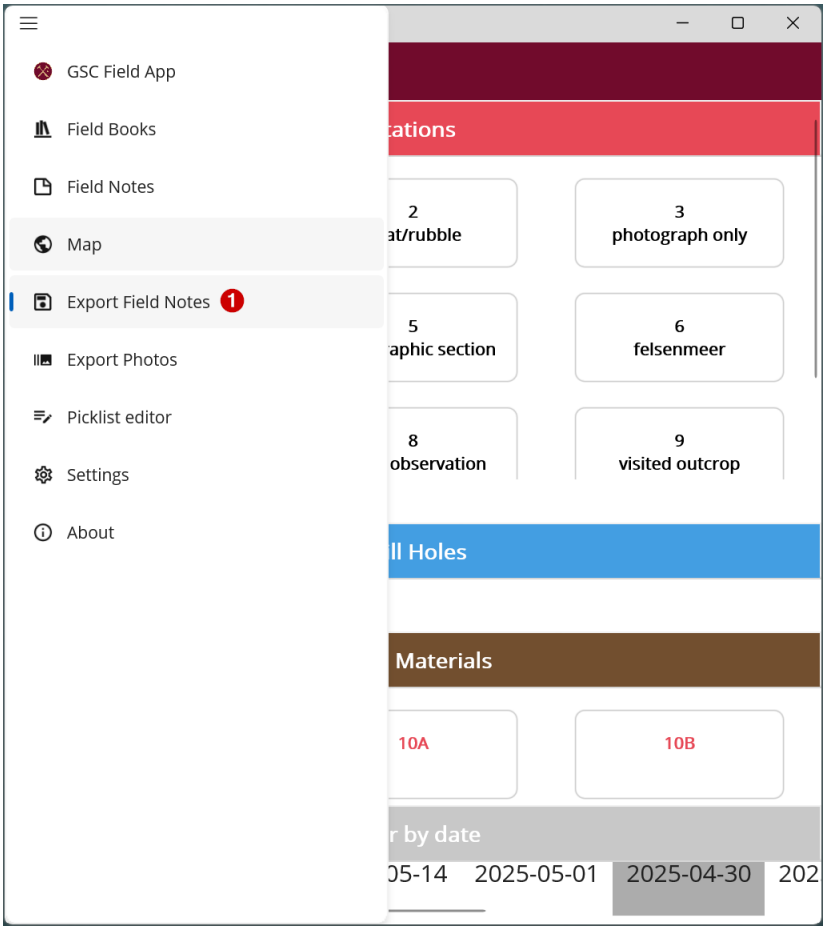


Map view: top and bottom navigation bars.

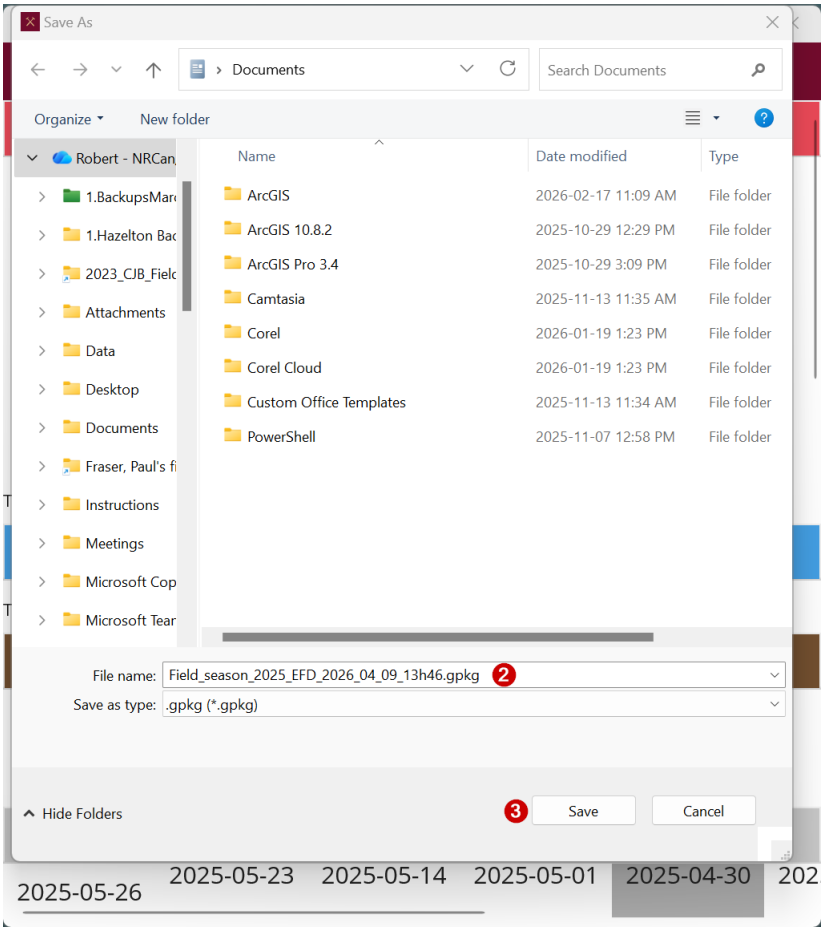
1. Main menu (Field Books, Field Notes, the Map View, Export Field Notes, Export Photos, Picklist Editor, Settings, About).
2. Tap to add a station record.
3. Tap to add a quick sample record.
4. Tap to add a quick photo record.
5. Tap to add a quick structure/pflow record.
6. Tap to add a waypoint.
7. Tap to manually create a location record from XY or lat/long.
8. Zoom in.
9. Zoom out.
10. Centre the map on the location.
11. Reorient the map direction (put north to the north).
12. Tap to make a distance measurement

13. Make a tap entry station (using the tapped location on the map)
14. Tap to draw an interpretation line.
15. Tap to add a URL of an online Web Mapping Server (WMS).
16. Tap to add a new layer.
17. Tap to see a list of all loaded layers and more options.
18. Tap to turn on/off GPS.

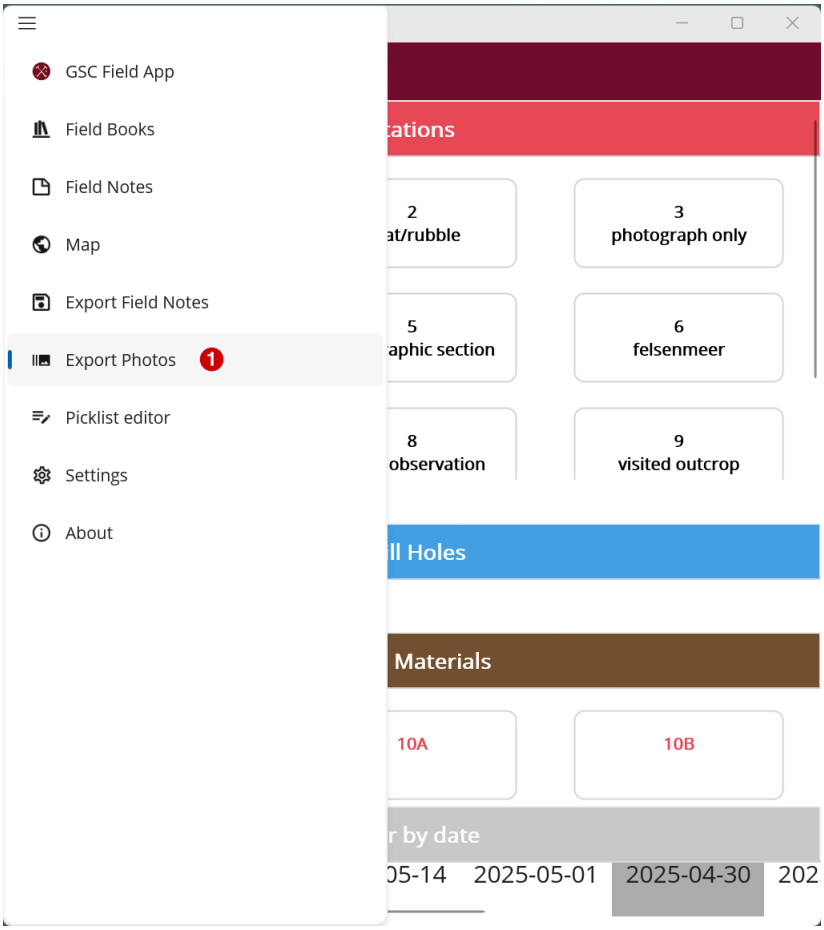
EXPORT FIELD NOTES / EXPORTER DES NOTES DE TERRAIN



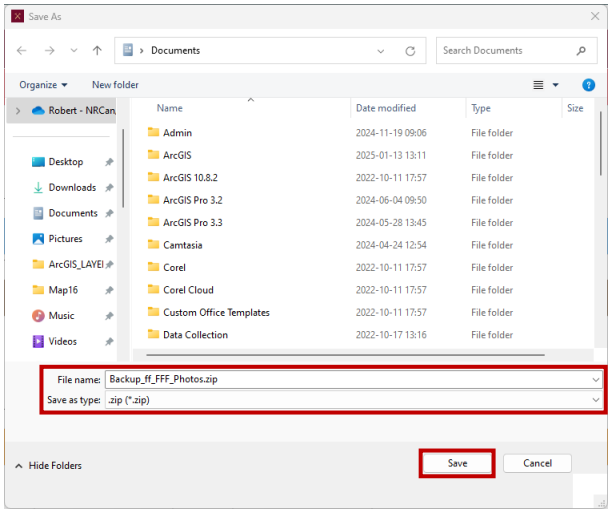
1. Tap here to export the field notes collected by the field device (as a GeoPackage)
2. Enter in the name of the backup file (.gpkg)
3. Tap the **Save** button


Save As
Documents
Search Documents
Organize
New folder
Robert - NRCan
1.BackupsMar
1.Hazelton Bac
2023_CJB_Fielc
Attachments
Data
Desktop
Documents
Fraser, Paul's fi
Instructions
Meetings
Microsoft Cop
Microsoft Tear
Name
Date modified
Type
ArcGIS
2026-02-17 11:09 AM
File folder
ArcGIS 10.8.2
2025-10-29 12:29 PM
File folder
ArcGIS Pro 3.4
2025-10-29 3:09 PM
File folder
Camtasia
2025-11-13 11:35 AM
File folder
Corel
2026-01-19 1:23 PM
File folder
Corel Cloud
2026-01-19 1:23 PM
File folder
Custom Office Templates
2025-11-13 11:34 AM
File folder
PowerShell
2025-11-07 12:58 PM
File folder
File name:
Field_season_2025_EFD_2026_04_09_13h46.gpkg
2
Save as type:
.gpkg (*.gpkg)
Hide Folders
3
Save
Cancel
2025-05-26
2025-05-23
2025-05-14
2025-05-01
2025-04-30

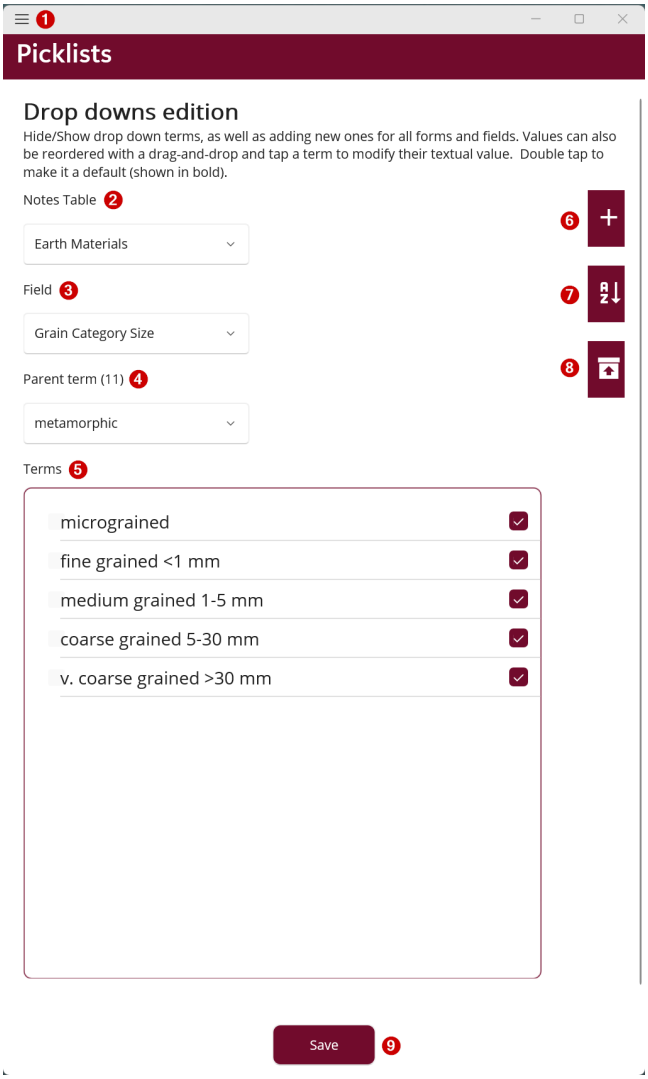
EXPORT PHOTOS / PHOTOS D'EXPORTATION



1. Tap here to export the photos collected by the field device (not an external camera)
2. Choose daily or all photos.
3. Enter in the name of the backup file (.zip)
4. Tap the **Save** button

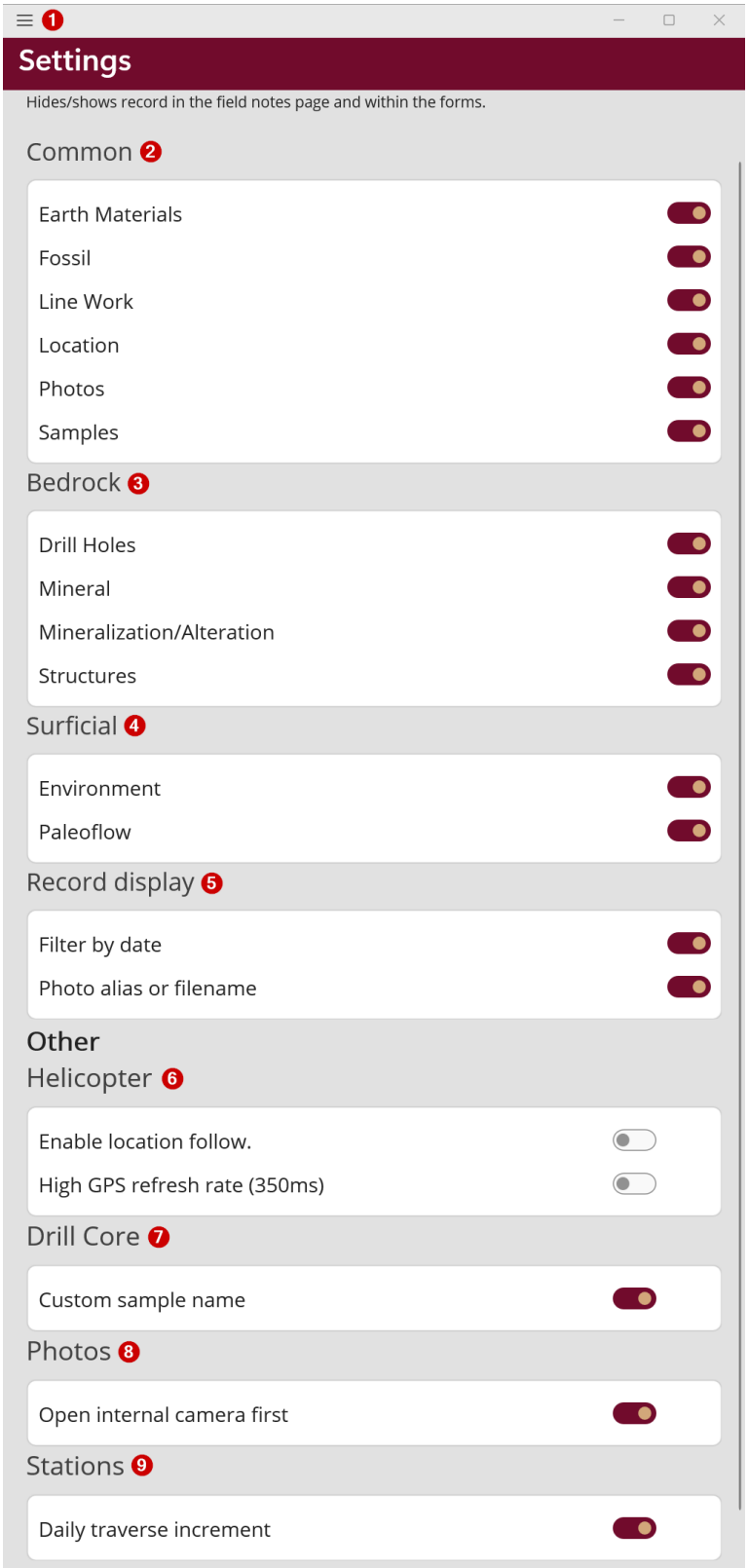


USING THE PICKLIST EDITOR / UTILISER L'ÉDITEUR DE LISTES DE SÉLECTION



1. Main Menu
2. The name of the table you wish to modify
3. The name of the field in the above table being modified
4. The name of the parent term being modified
5. The terms contained under the parent term
6. Tap to add a new picklist term
7. Tap to sort in ascending order (alphabetical)
8. Tap to import picklist from another geopackage
9. Save the current picklists

SETTINGS / PARAMÈTRES



1. Main Menu
2. Settings for the tables that are common between both Bedrock and Surficial
3. Settings for Bedrock only
4. Settings for Surficial only
5. Settings affecting the display of records
6. Settings related to using the GSC Field App in a helicopter
7. Settings related to Drill Holes
8. Settings related to Photos
9. Settings related to Stations

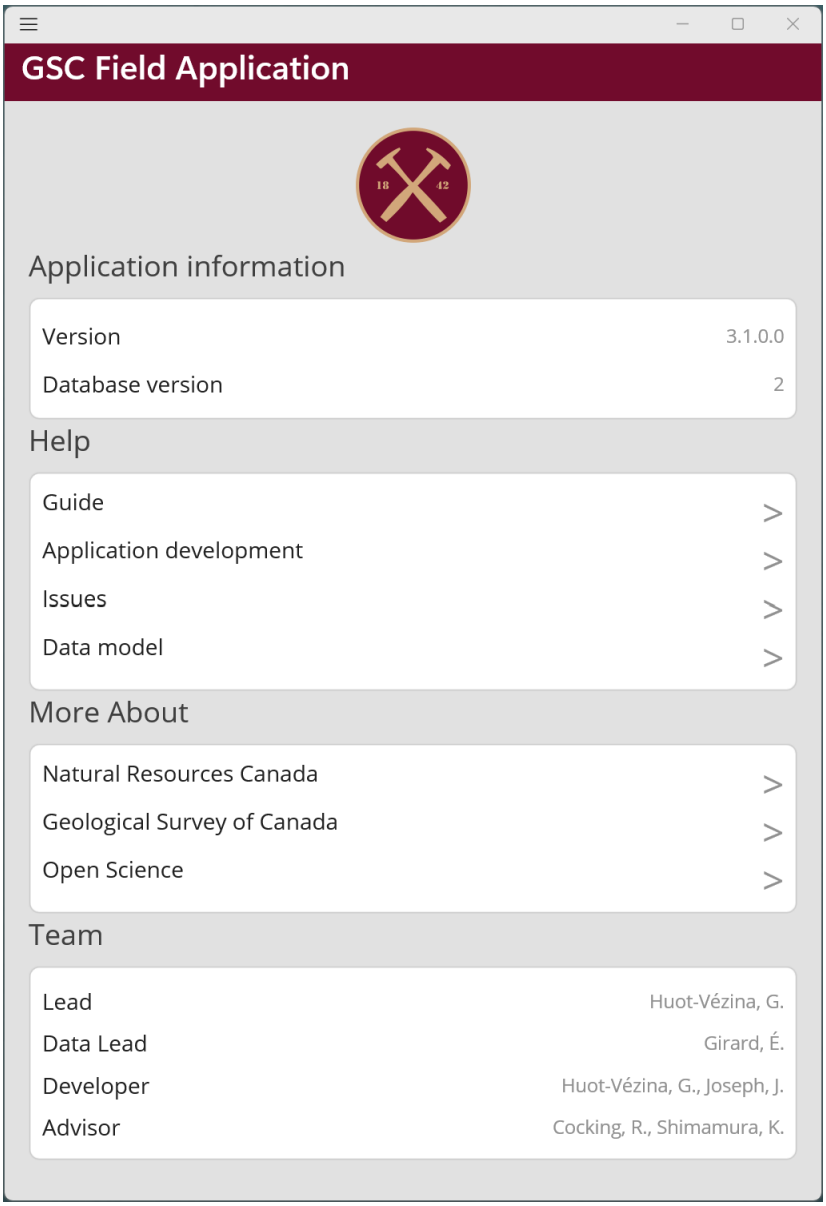
The Settings – allows you to turn on/off certain tables of data collection. For example, if you know you don't have any interest in collecting drill hole data or mineralization/alteration, you can turn those tables off.

There are some settings that allow you to optimize the device in a helicopter setting, where you need a faster refresh rate on the GPS since the helicopter is travelling so fast.

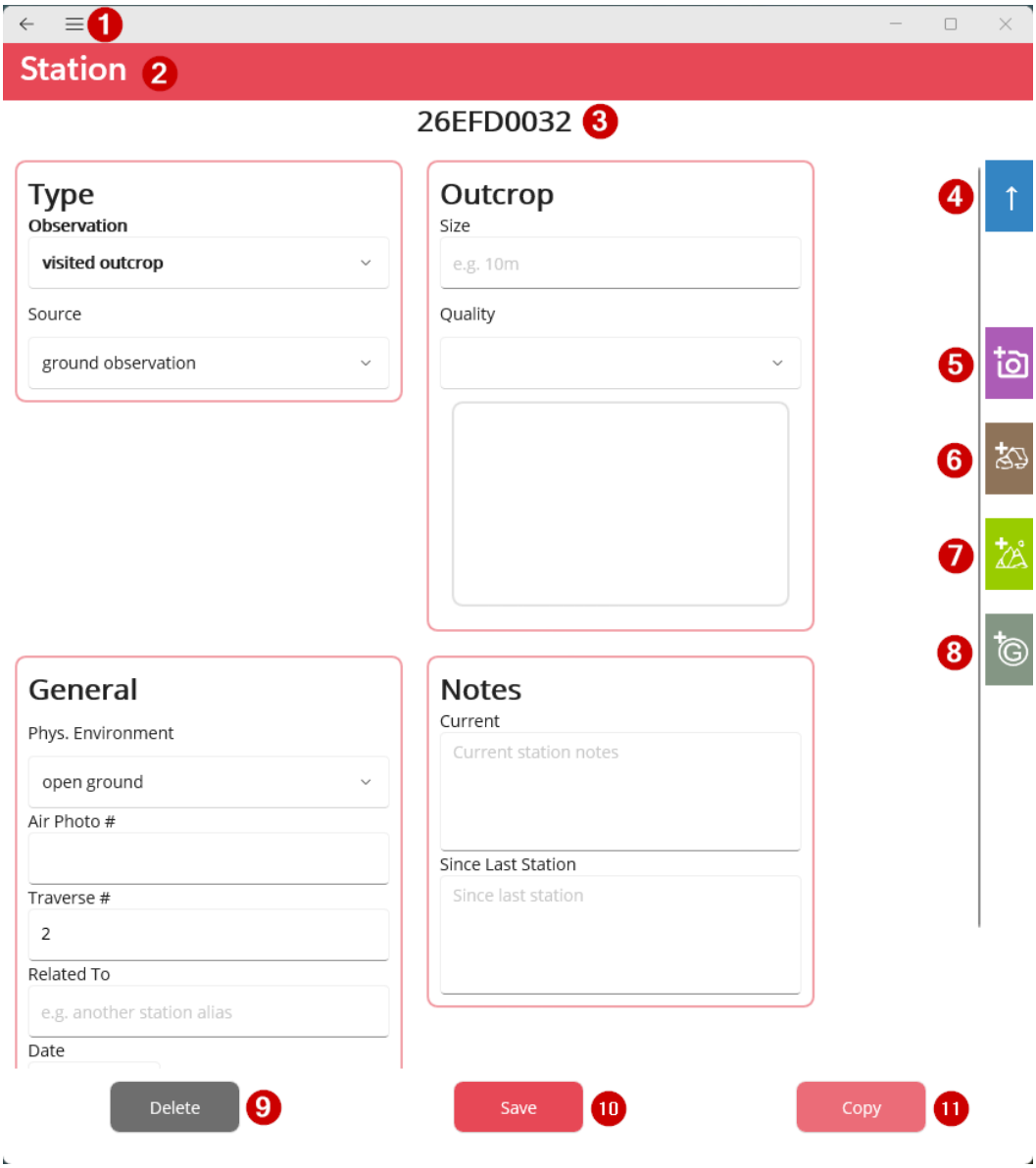
Additionally, you can customize a sample name, open the internal camera by default and customize the daily traverse increment.

ABOUT / À PROPOS

The About page lists the App version and Database version, the main developers and, more importantly, provides links to help (if the user is connected to the Internet).



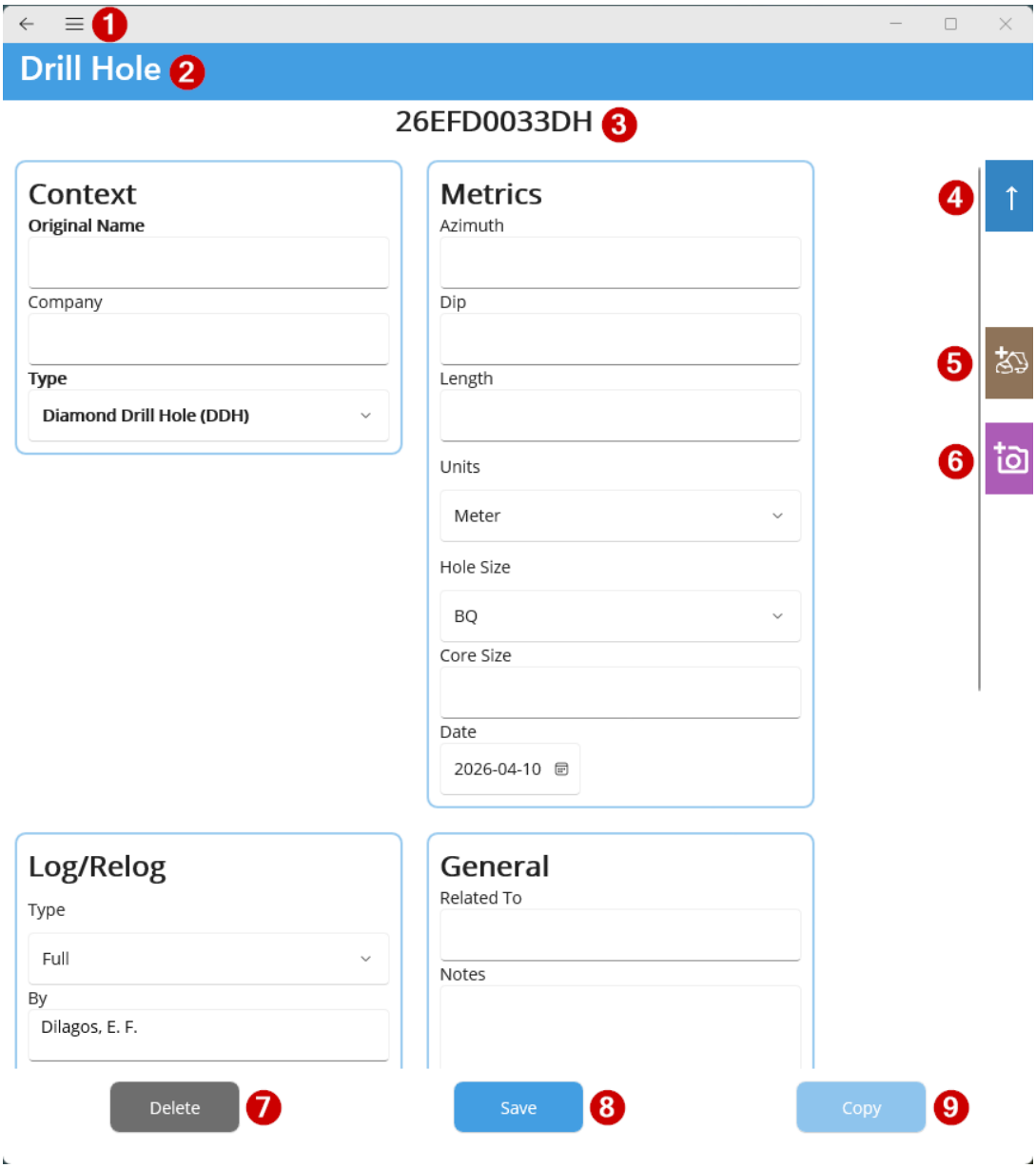
STATION NOTES / NOTES DE STATION



1. Main Menu
2. Station record
3. Station Name
4. Tap to navigate to parent record
5. Tap to add one or many photos
6. Tap to add a new earth material
7. Tap to add a new environment
8. Tap to add a new mineralization/alteration
9. Delete this record
10. Save any edits made to this record
11. Copy this record

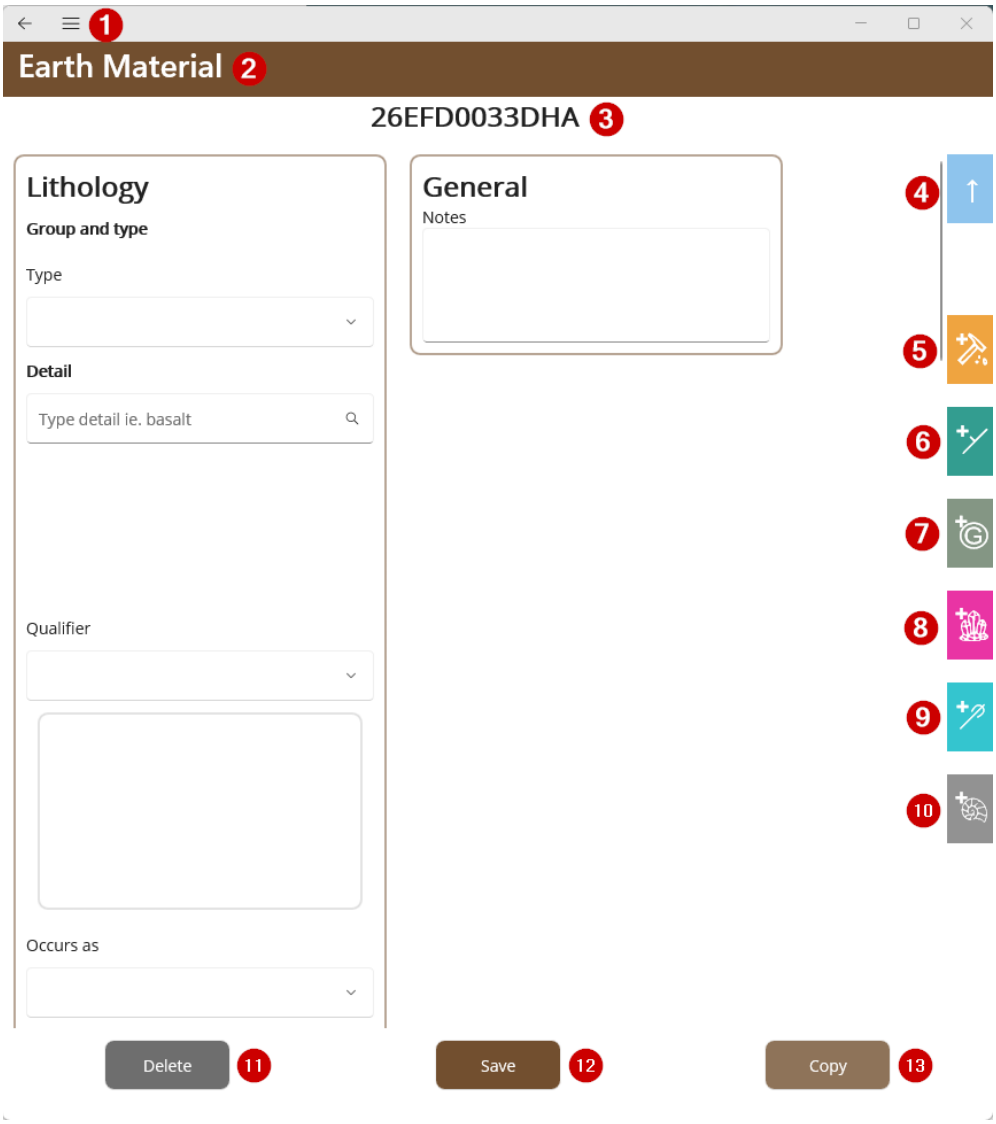
DRILL HOLE NOTES / NOTES SUR LES TROUS DE FORAGE

Note: add a drill hole by first adding a Manual XY entry from the map page then tap the blue Drill Hole +.



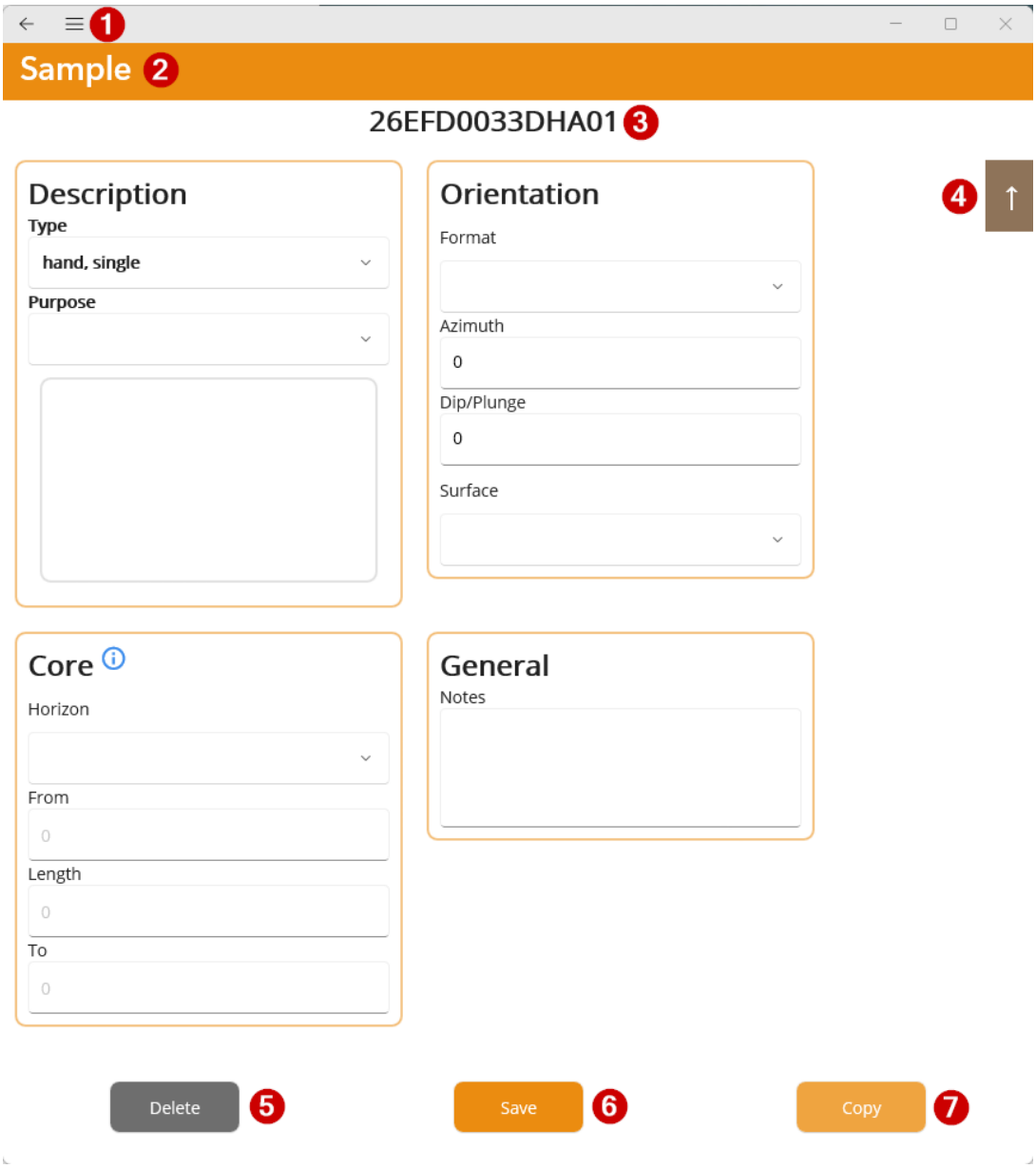
1. Main Menu
2. Drill Hole record
3. Drill Hole Name
4. Tap to navigate to parent record
5. Tap to add a new earth material
6. Tap to add one or many photos
7. Delete this record
8. Save any edits made to this record
9. Copy this record

EARTH MATERIAL NOTES / NOTES SUR LES MATÉRIAUX TERRESTRES



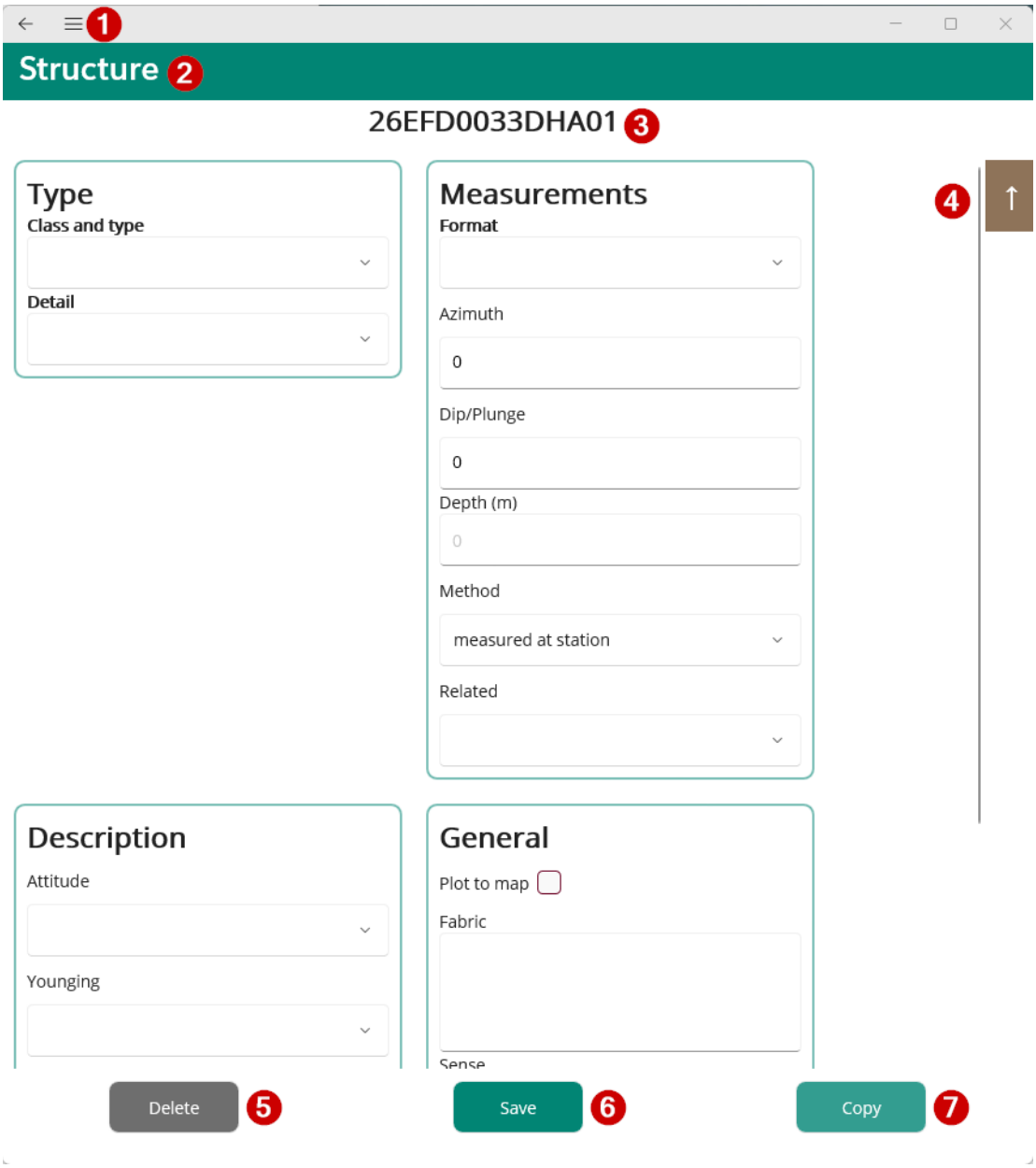
1. Main Menu
2. Earth Material record
3. Earth Material Name
4. Tap to navigate to parent record
5. Tap to add a new sample
6. Tap to add a new structure
7. Tap to add a new mineralization/alteration
8. Tap to add a new mineral
9. Tap to add a new paleoflow
10. Tap to add a new fossil
11. Delete this record
12. Save any edits made to this record
13. Copy this record

SAMPLE NOTES / EXEMPLES DE NOTES



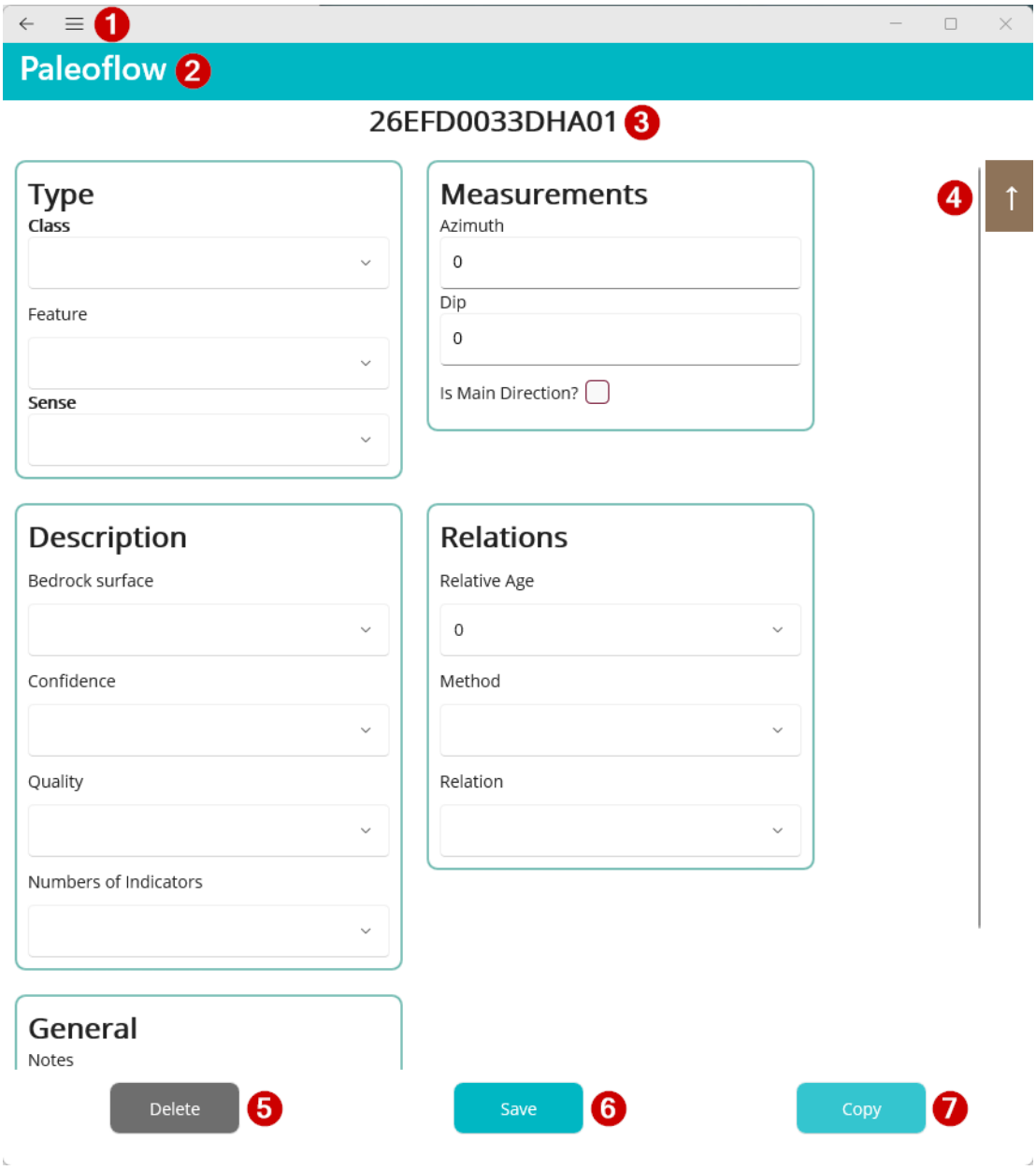
1. Main Menu
2. Sample record
3. Sample Name
4. Tap to navigate to parent record
5. Delete this record
6. Save any edits made to this record
7. Copy this record

STRUCTURE NOTES / NOTES DE STRUCTURE



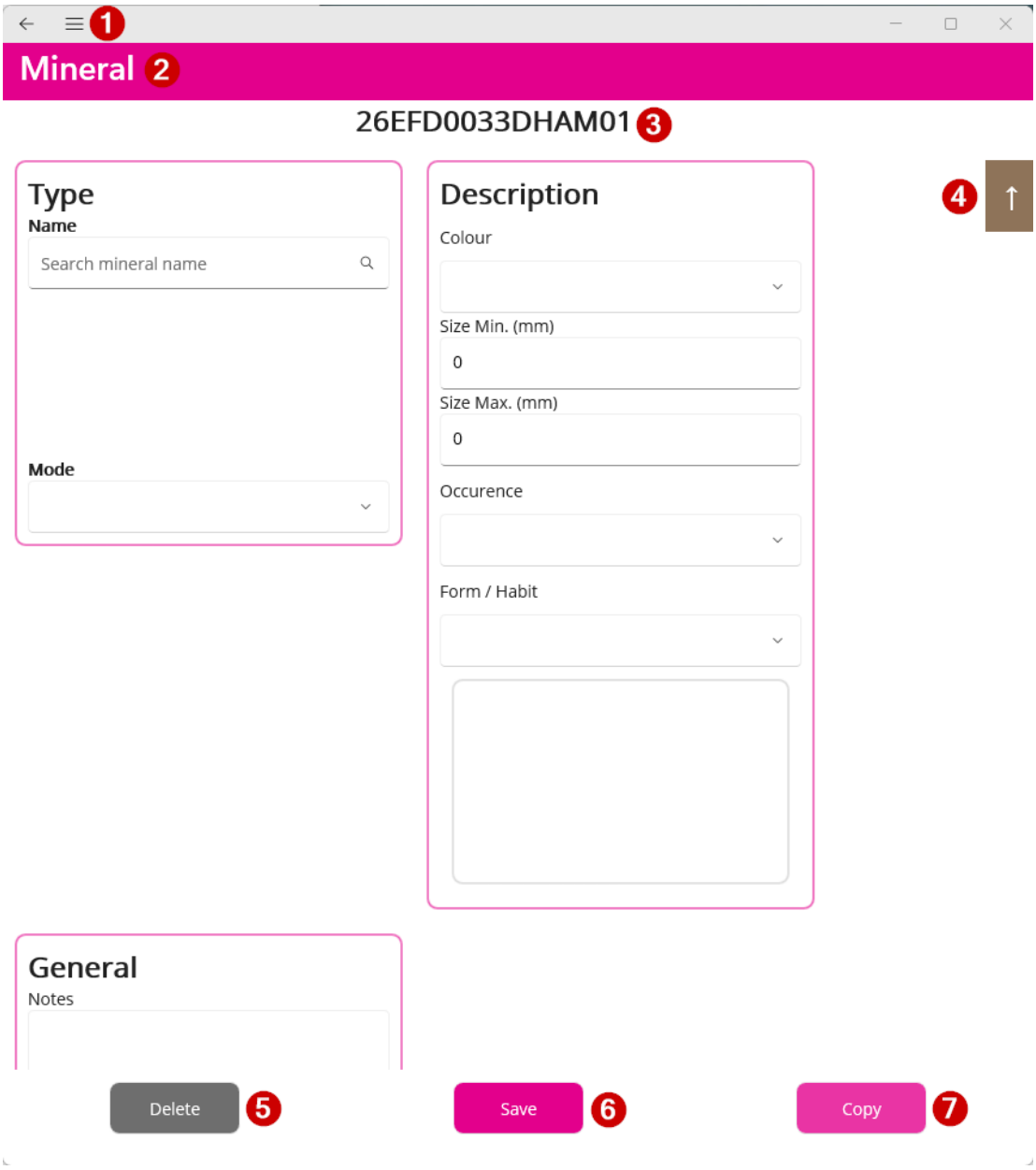
1. Main Menu
2. Structure record
3. Structure Name
4. Tap to navigate to parent record
5. Delete this record
6. Save any edits made to this record
7. Copy this record

PALEOFLOW NOTES / NOTES SUR LE PALEOFLOW



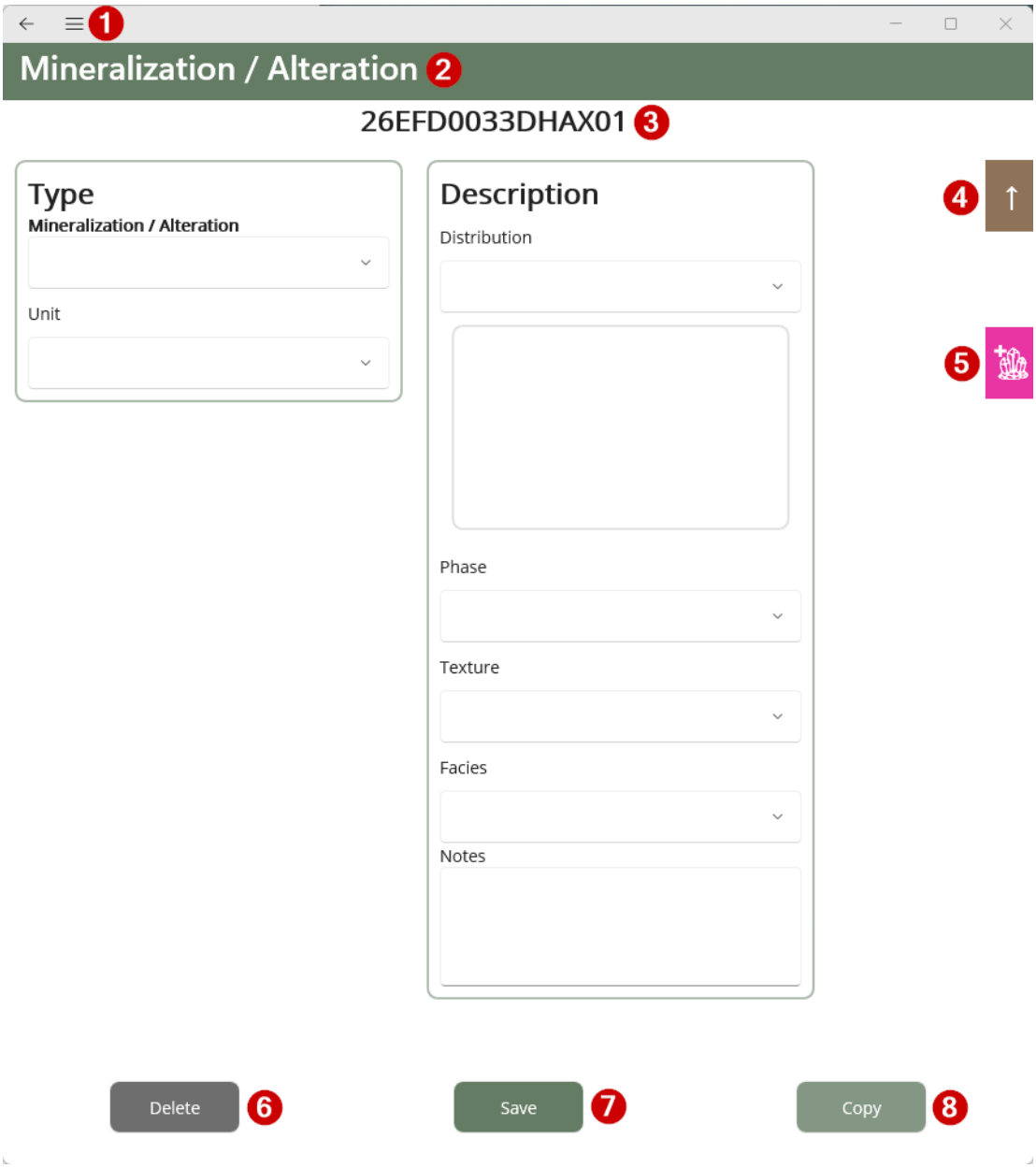
1. Main Menu
2. Paleoflow record
3. Paleoflow Name
4. Tap to navigate to parent record
5. Delete this record
6. Save any edits made to this record
7. Copy this record

MINERAL NOTES / NOTES SUR LE MINÉRALES



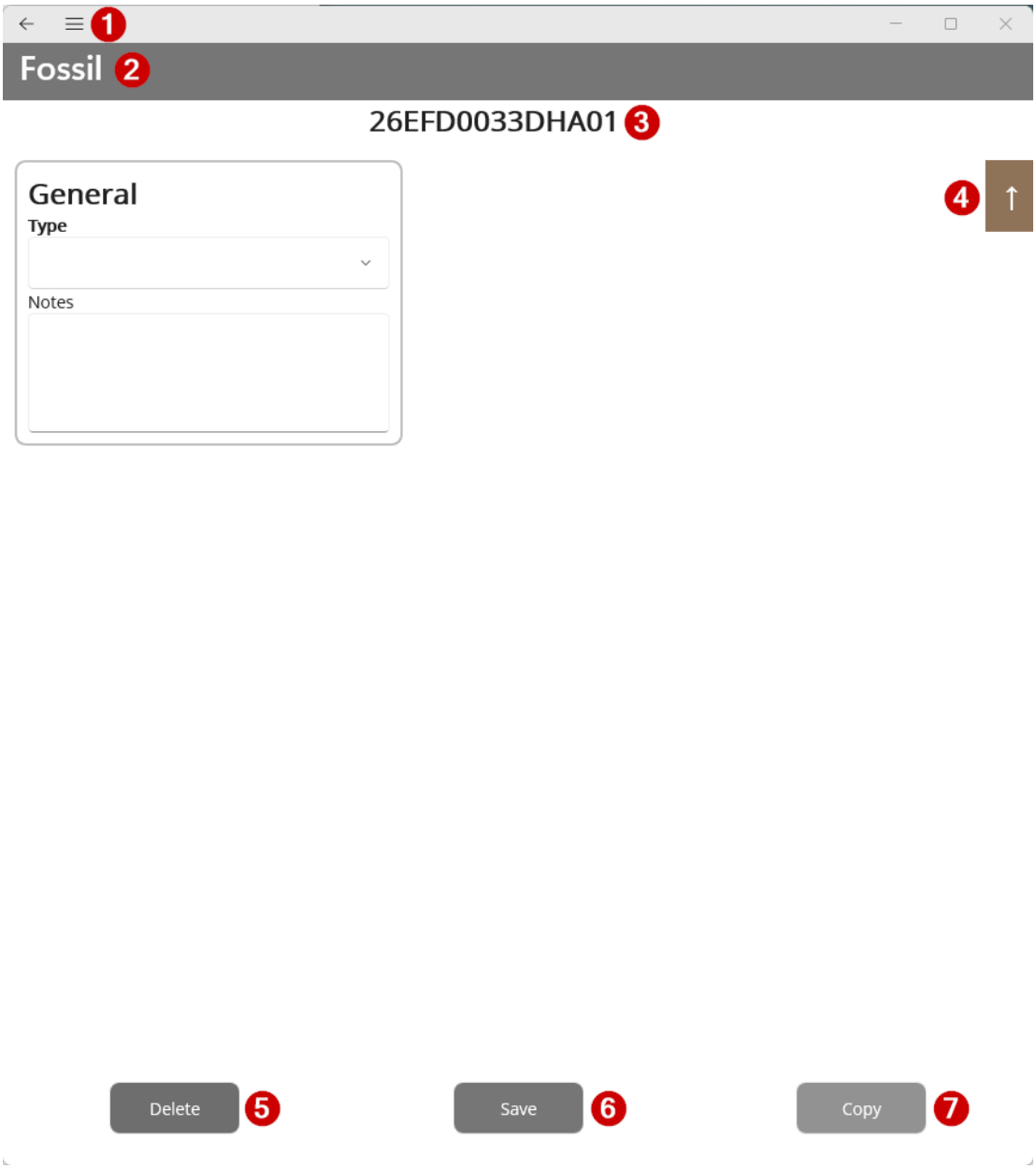
1. Main Menu
2. Mineral record
3. Mineral Name
4. Tap to navigate to parent record
5. Delete this record
6. Save any edits made to this record
7. Copy this record

MINERALIZATION-ALTERATION NOTES / NOTES SUR LA MINÉRALISATION-ALTERATION



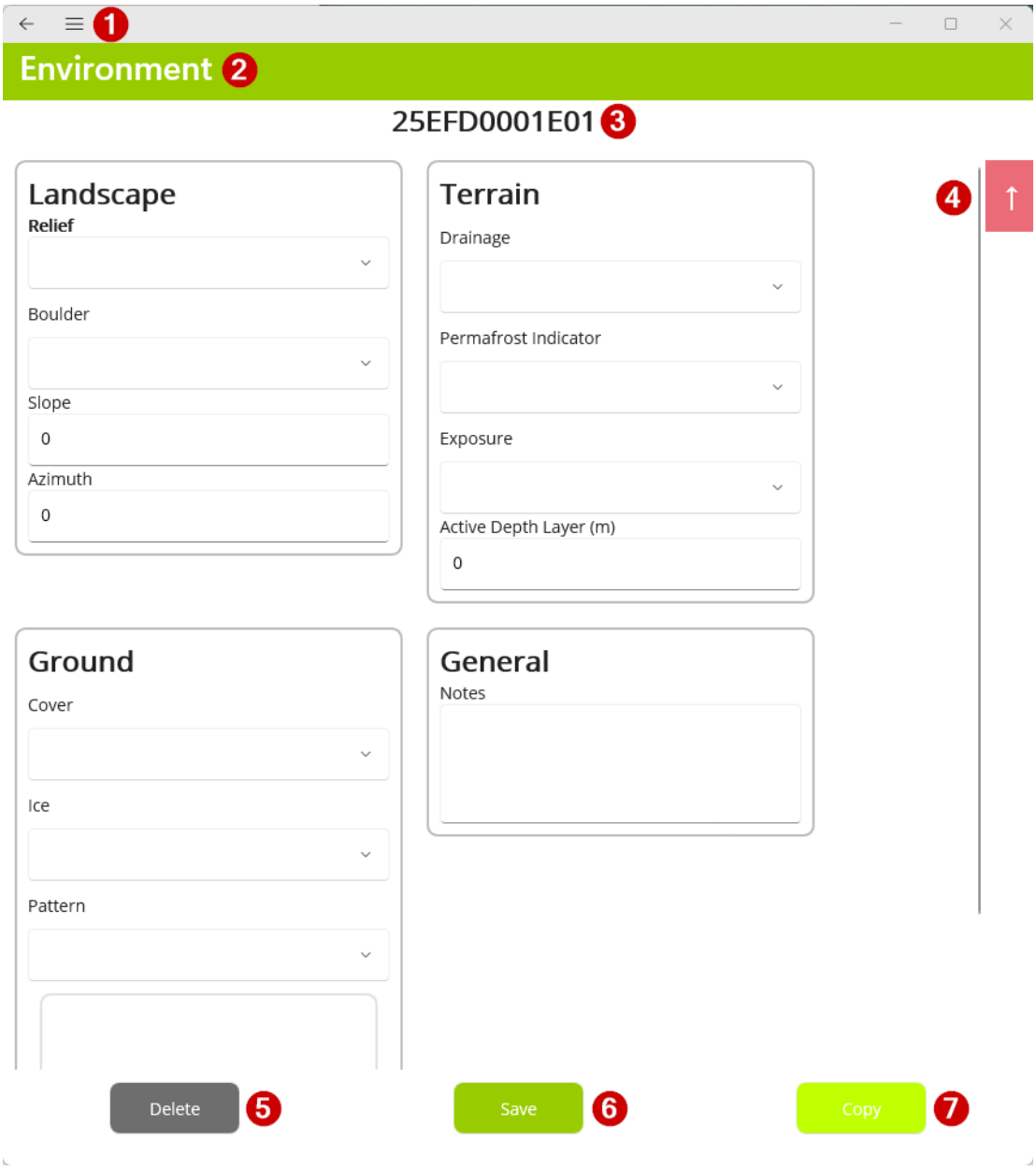
1. Main Menu
2. Mineralization/Alteration record
3. Mineralization/Alteration Name
4. Tap to navigate to parent record
5. Tap to add a new mineral
6. Delete this record
7. Save any edits made to this record
8. Copy this record

FOSSIL NOTES / NOTES SUR LES FOSSILES



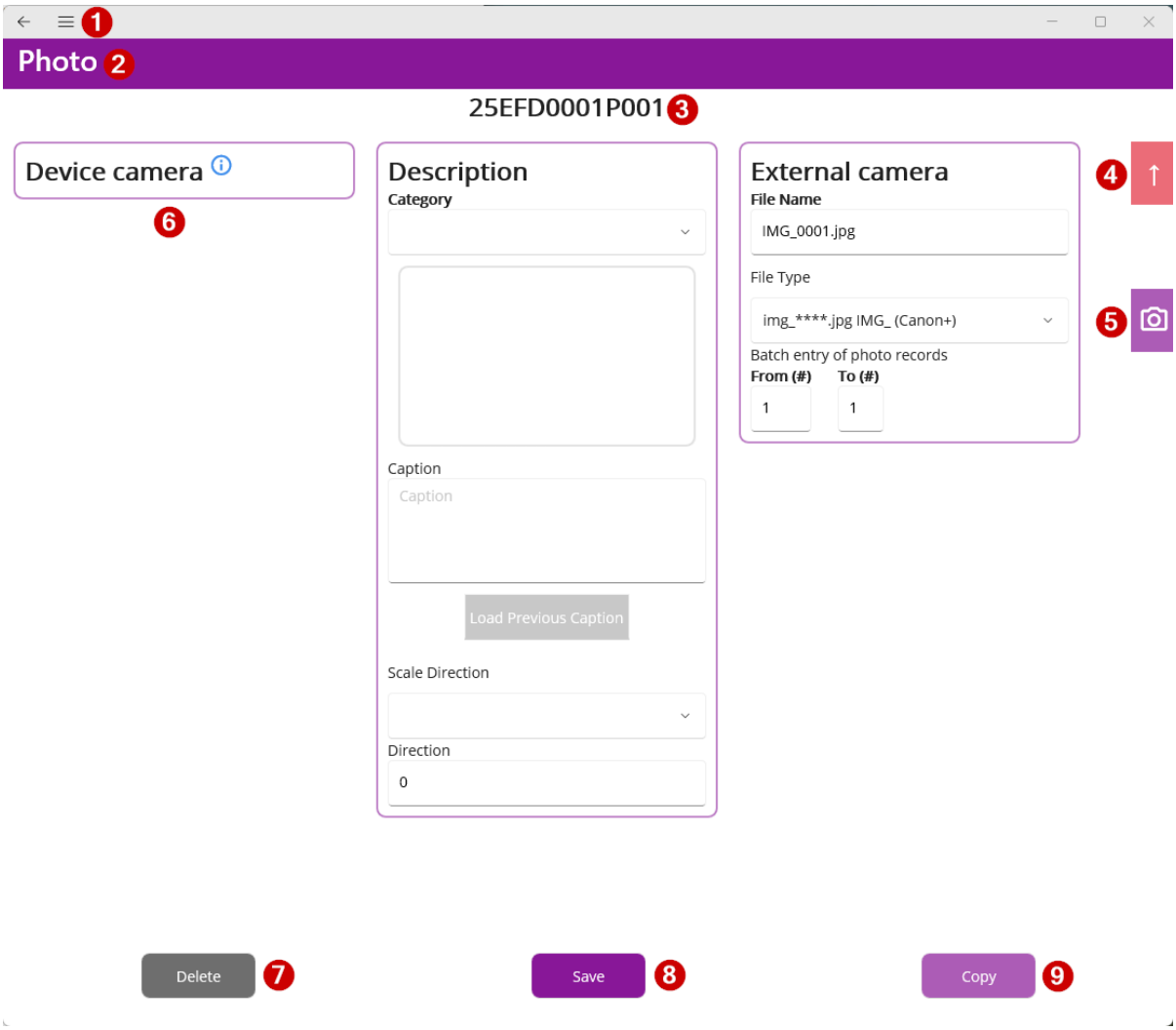
1. Main Menu
2. Fossil record
3. Fossil Name
4. Tap to navigate to parent record
5. Delete this record
6. Save any edits made to this record
7. Copy this record

ENVIRONMENT NOTES / NOTES SUR L'ENVIRONNEMENT



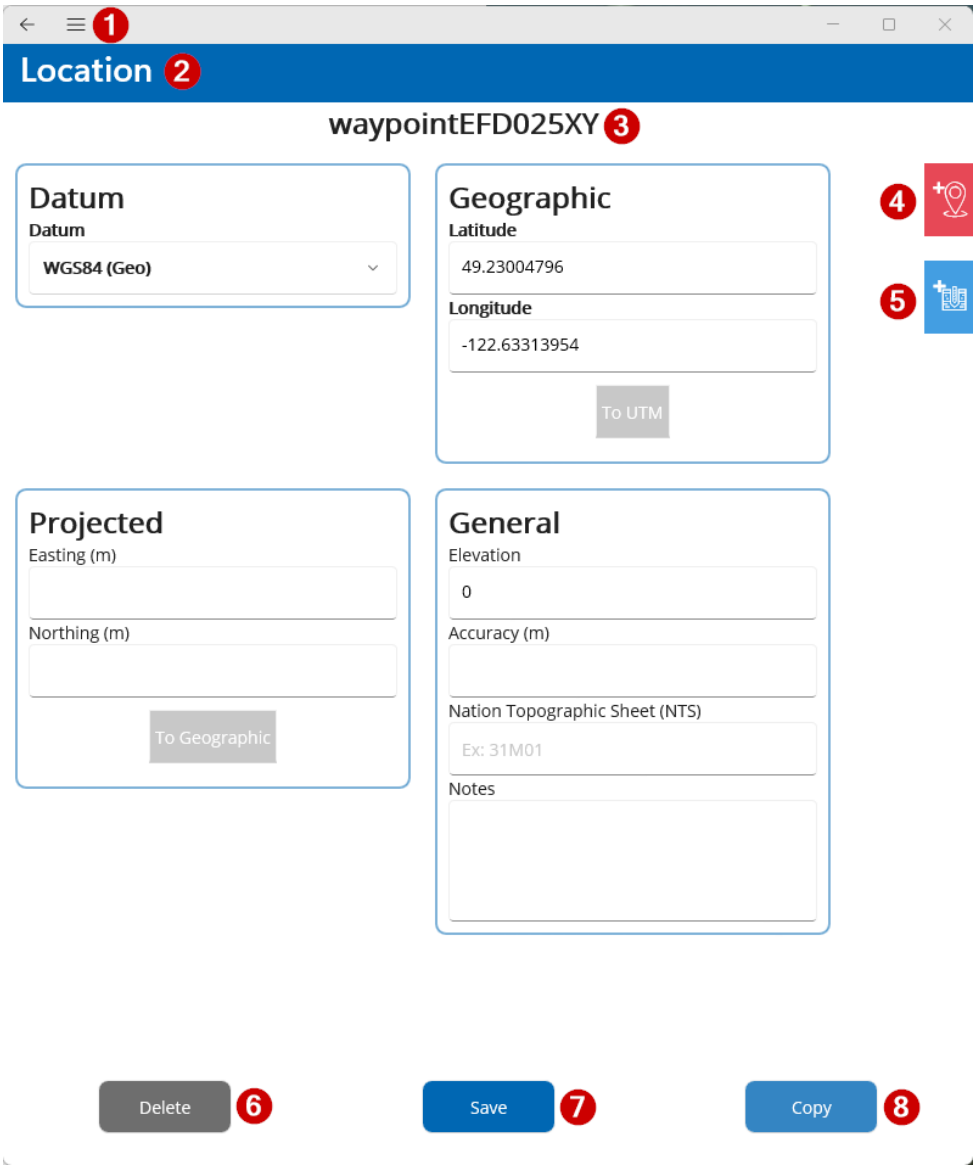
1. Main Menu
2. Environment record
3. Environment Name
4. Tap to navigate to parent record
5. Delete this record
6. Save any edits made to this record
7. Copy this record

PHOTOS NOTES / NOTES DE PHOTOS



1. Main Menu
2. Photos record
3. Photos Name
4. Tap to navigate to parent record
5. Take a snapshot with device
6. A snapshot taken with the device
7. Delete this record
8. Save any edits made to this record
9. Copy this record

LOCATION NOTES / NOTES DE LOCALISATION



1. Main Menu
2. Location record
3. Location Name
4. Tap to add a new station record
5. Tap to add a new drill hole record
6. Delete this record
7. Save any edits made to this record
8. Copy this record

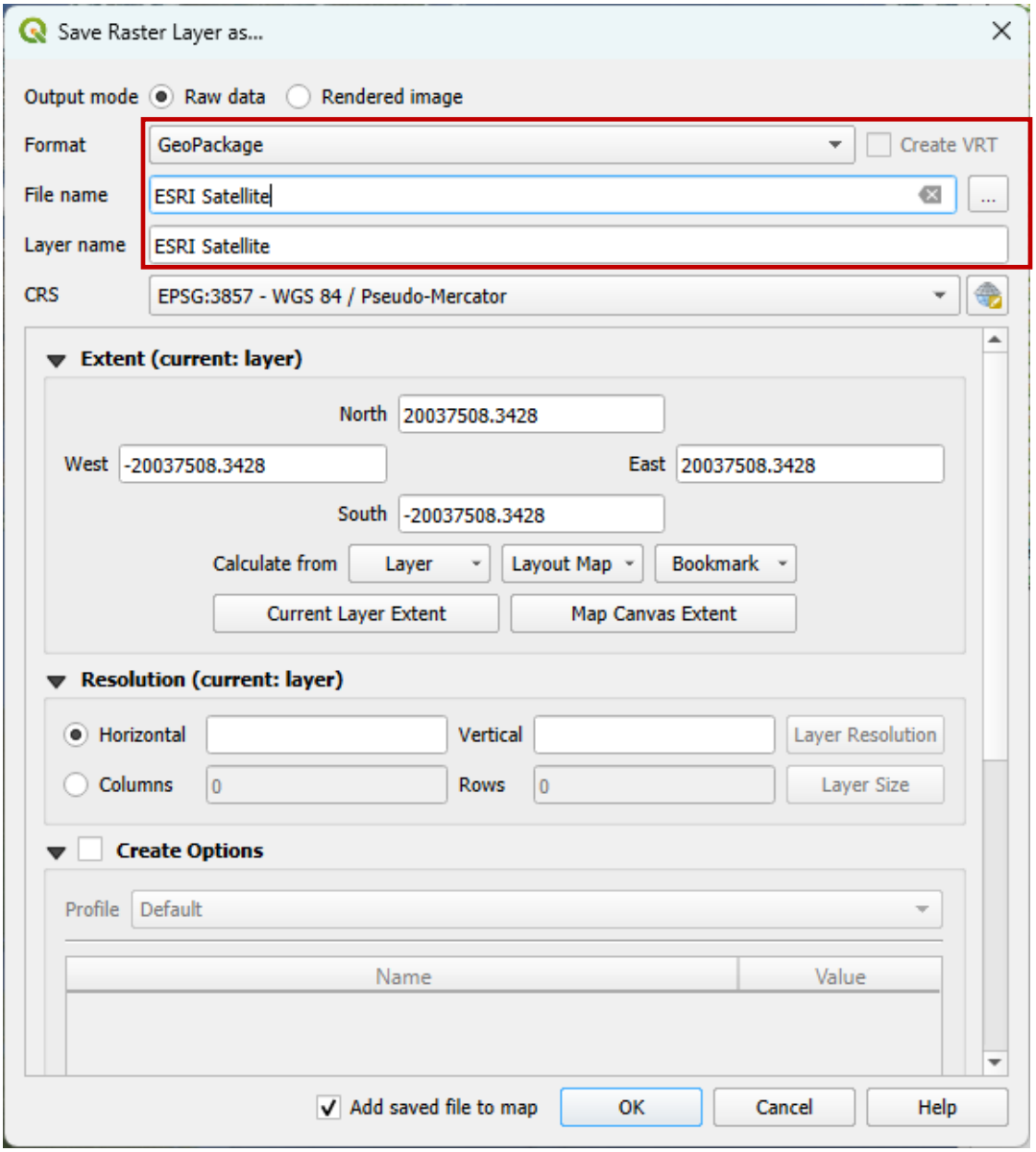
USING GEOPACKAGES / UTILISER LES GÉOPACKAGES

Like MBTiles, you can create and use geopackage-formatted files in the GSC Field App. Additionally, if you create the geopackage in QGIS, you can include the same symbololgy used in QGIS in the Field App

NOTE: Only two projections are used in the GSC Field App. If you wish to create a geopackage, the projection of the data must be either geographic WGS 84 (EPSG: 4326) or WGS 84 / Pseudo-Mercator (EPSG: 3857).

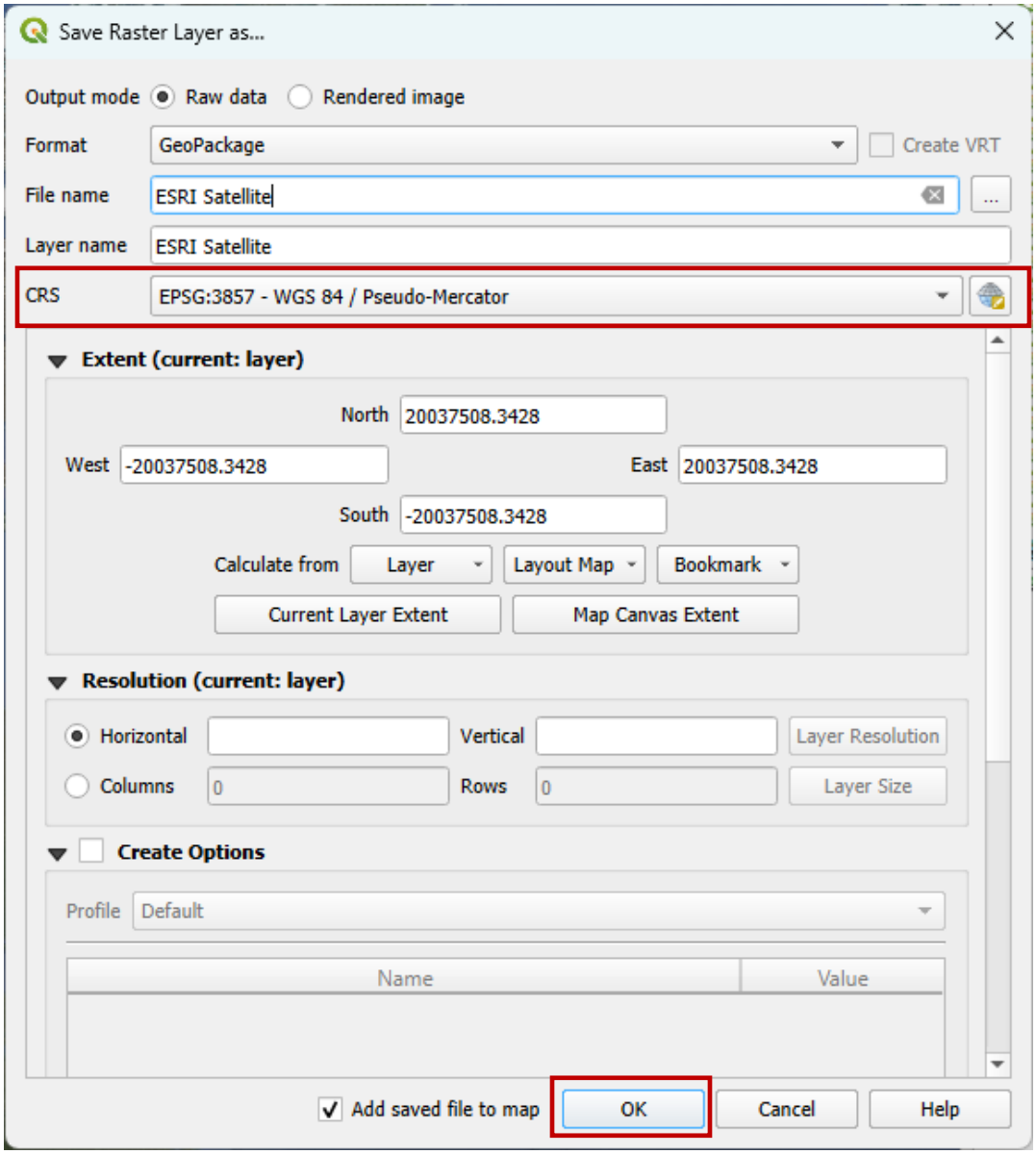
CREATING GEOPACKAGES IN QGIS / CRÉATION DE GÉOPACKAGES DANS QGIS

1. In the QGIS Table of Contents, right-click on the map layer you wish to be used in a geopackage and select **Export > Save As…**
2. Set the Format to **GeoPackage** and name the GeoPackage



3.

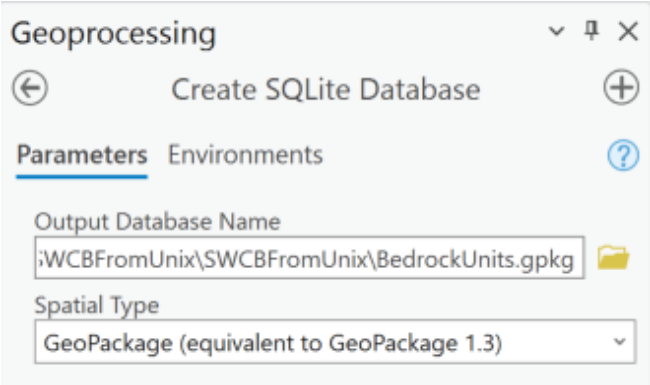
4. Set the CRS projection to either EPSG: 3857 or EPSG: 4326 (the only projections supported by the GSC Field App at this time)
5. Click OK to export the data


Save Raster Layer as...
Output mode
Raw data
Rendered image
Format
GeoPackage
Create VRT
File name
ESRI Satellite
Layer name
ESRI Satellite
CRS
EPSG:3857 - WGS 84 / Pseudo-Mercator
Extent (current: layer)
North
20037508.3428
West
-20037508.3428
East
20037508.3428
South
-20037508.3428
Calculate from
Layer
Layout Map
Bookmark
Current Layer Extent
Map Canvas Extent
Resolution (current: layer)
Horizontal
Vertical
Layer Resolution
Columns
0
Rows
0
Layer Size
Create Options
Profile
Default
Name
Value
Add saved file to map
OK
Cancel
Help

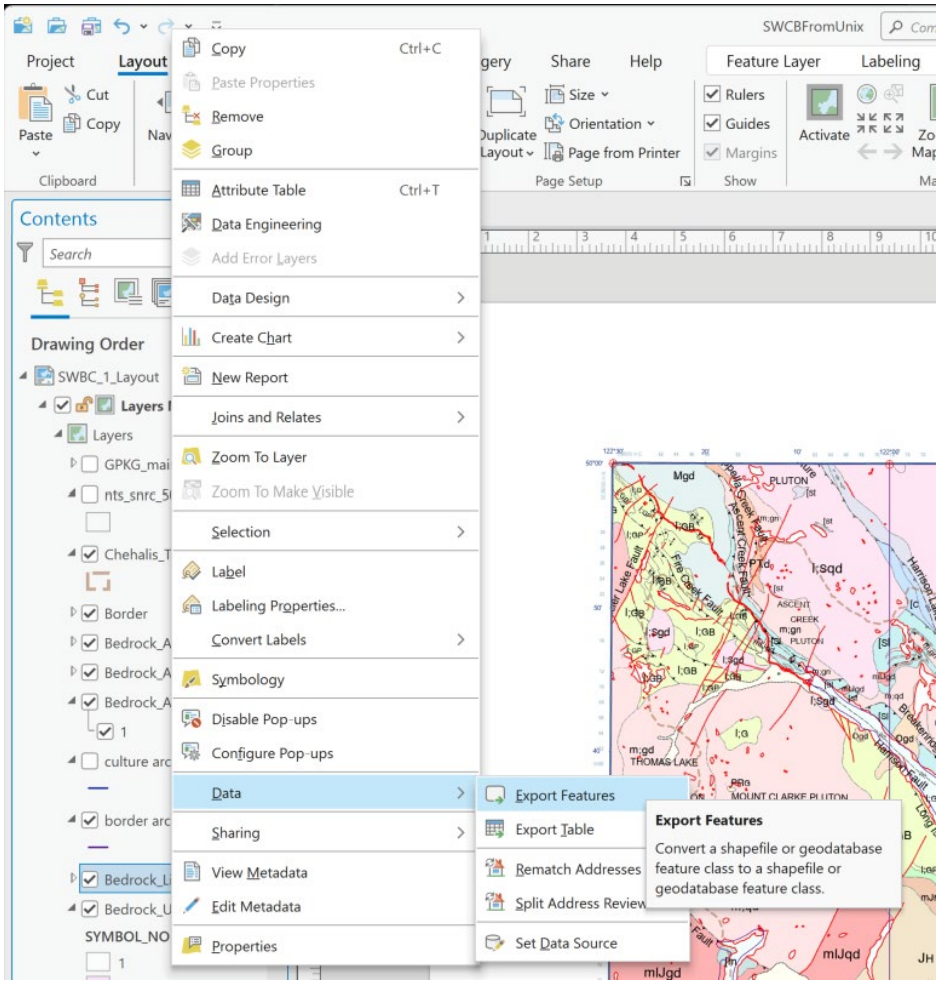
CREATING GEOPACKAGES IN ARCGIS PRO / CRÉATION DE GÉOPACKAGES DANS ARCGIS PRO

ArcGIS Pro works with data differently than QGIS. It requires the data be in an SQLite database

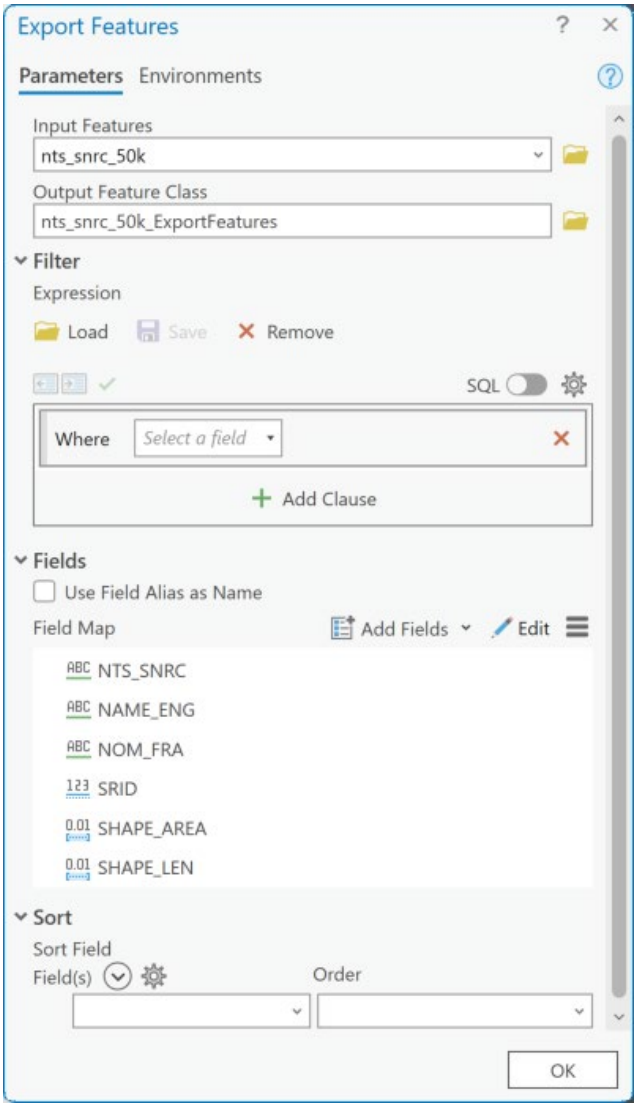
1. Create an SQLite database with the **Create SQLite Database** command.

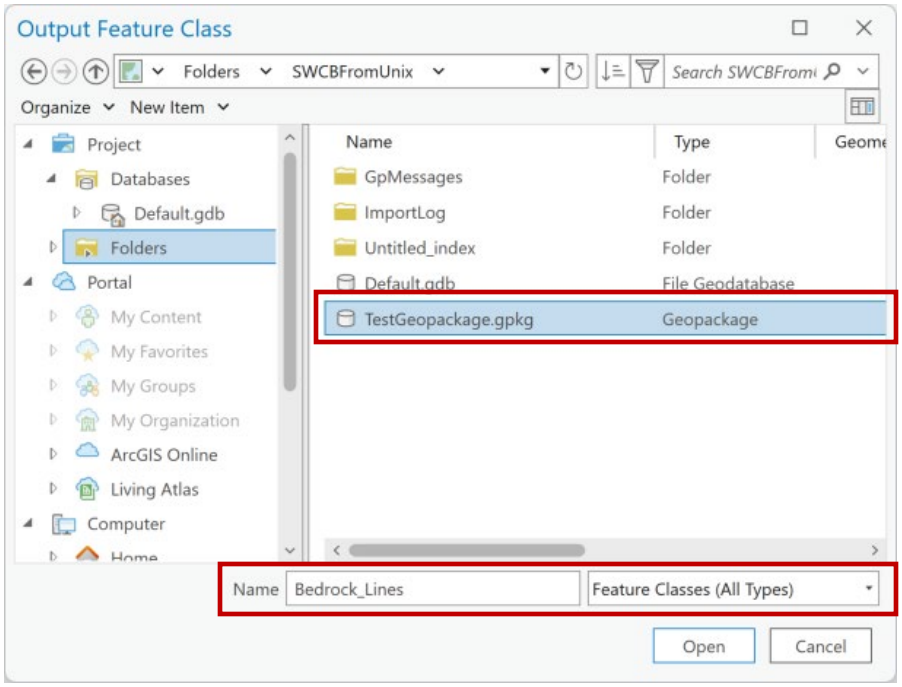


2. Ensure that you set the Spatial Type to 'GeoPackage (equivalent to GeoPackage 1.3)'. The file name will change to a .gpkg
3. Click on the **Run** button to create the geopackage
4. In the Table of Contents, right-click on the layer you want to add to the geopackage and navigate to **Data > Export Features**



5. Navigate to the location of the geopackage that you setup in Step 1 and enter the name of the layer.

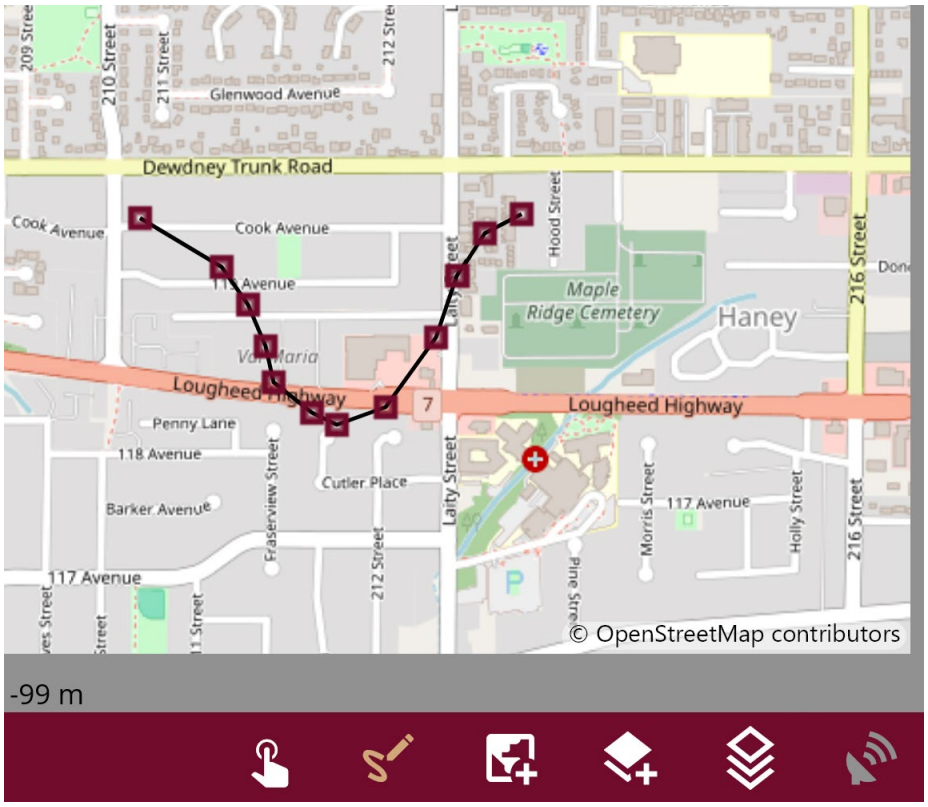

Export Features
Parameters
Environments
Input Features
nts_snrc_50k
Output Feature Class
nts_snrc_50k_ExportFeatures
Filter
Expression
Load
Save
Remove
SQL
Where
Select a field
Add Clause
Fields
Use Field Alias as Name
Field Map
Add Fields
Edit
NTS_SNRC
NAME_ENG
NOM_FRA
SRID
SHAPE_AREA
SHAPE_LEN
Sort
Sort Field
Field(s)
Order
OK


Output Feature Class
Folders
SWCBFromUnix
Search SWCBFrom
Organize
New Item
Project
Databases
Default.gdb
Folders
Portal
My Content
My Favorites
My Groups
My Organization
ArcGIS Online
Living Atlas
Computer
Home
Name
Type
Geome
GpMessages
Folder
ImportLog
Folder
Untitled_index
Folder
Default.gdb
File Geodatabase
TestGeopackage.gpkg
Geopackage
Name
Bedrock_Lines
Feature Classes (All Types)
Open
Cancel

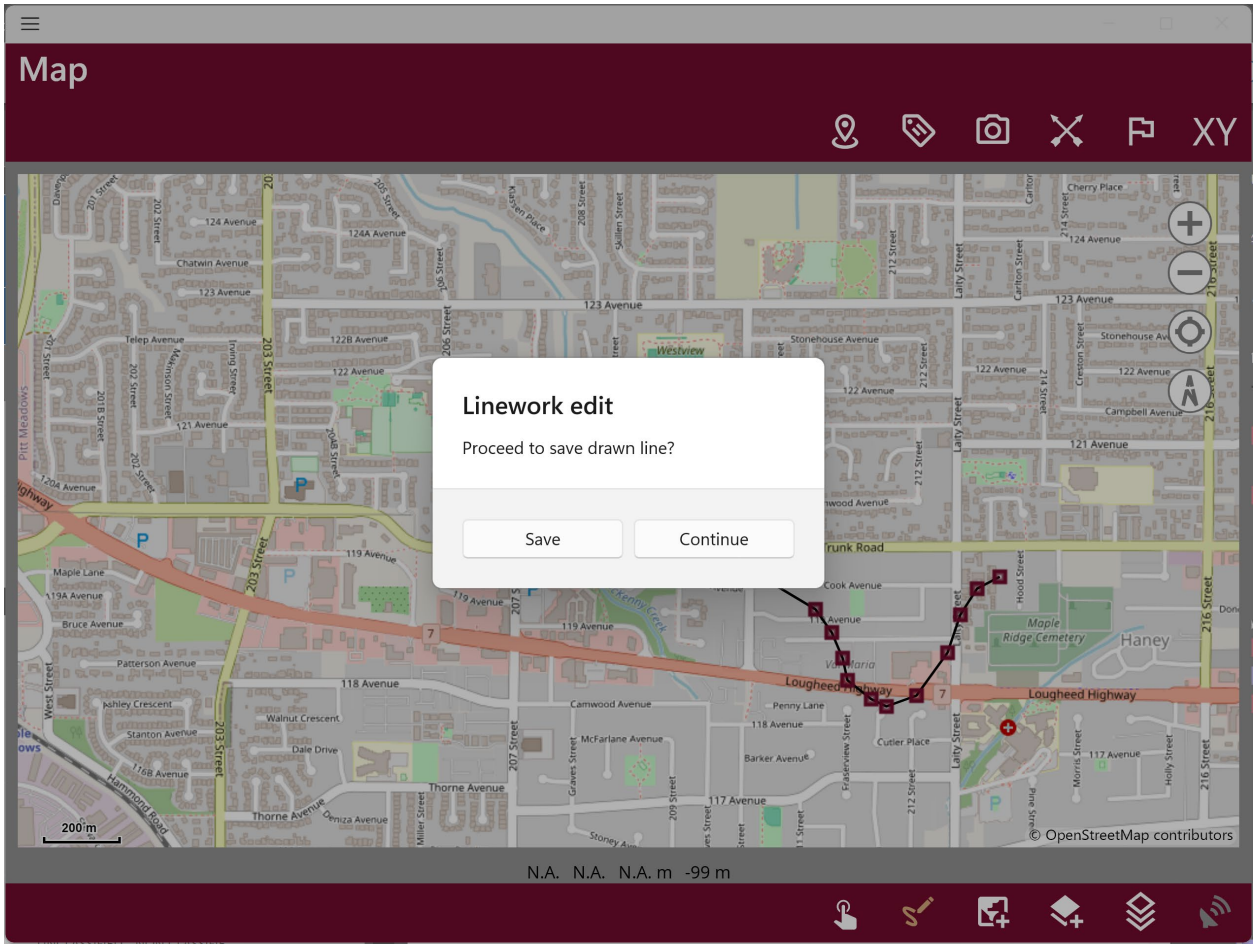
USING THE INTERPRETATION LINE FEATURE / L'UTILISATION DE LA LIGNE D'INTERPRÉTATION

Often, a field geologist will identify a linear feature that they would like to draw on the map. This can include contacts, faults, folds

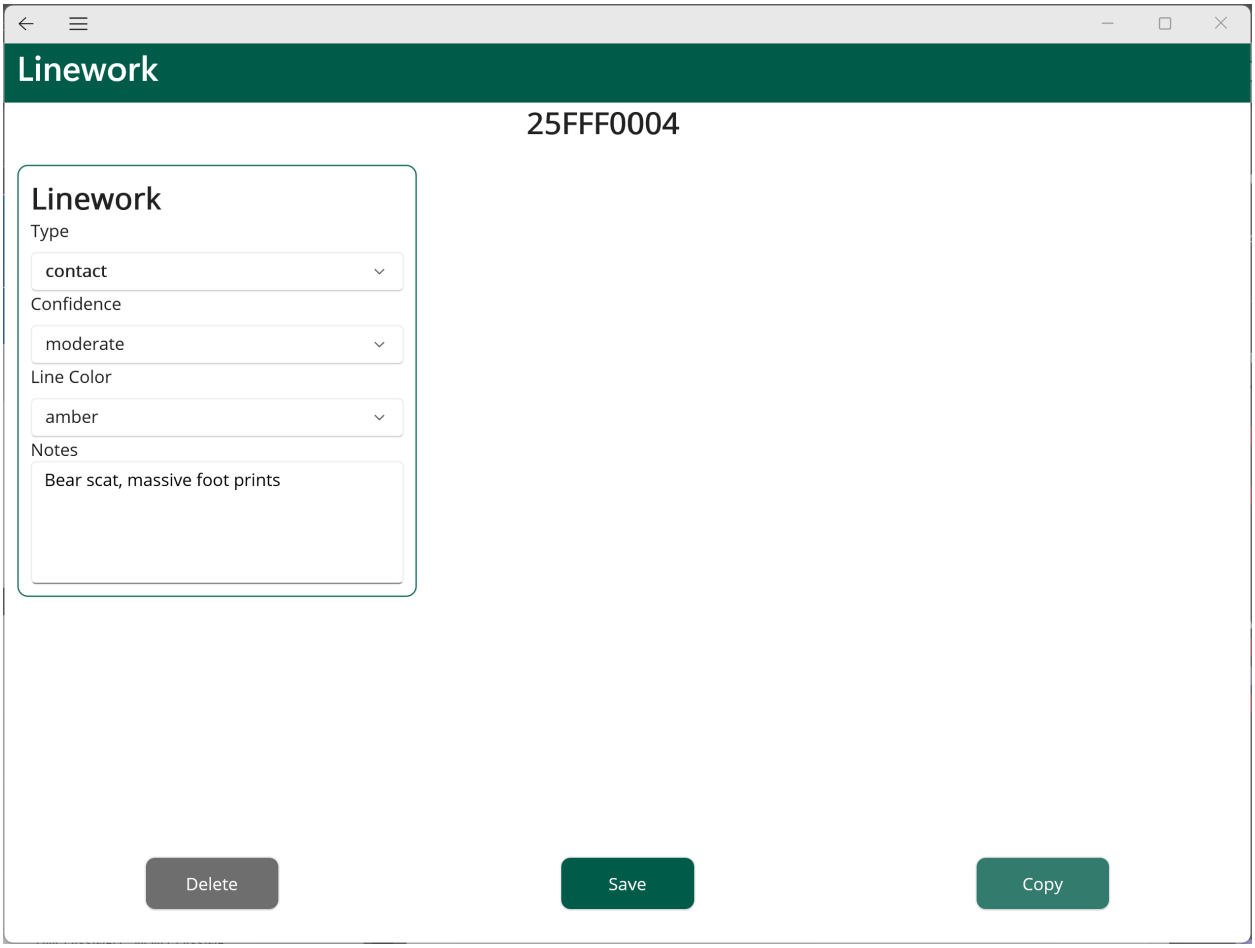
1. Click on the 'Tap to draw an interpretation line' button.
2. The button will change to show it is in 'draw mode' .
3. Tap on the screen where you want to start the line, then point to all the vertices you wish to add.



4. Once the line is completed, either double tap or hold the finger position for 2 seconds and lift your finger.



5. The Line attribute menu will appear. Choose the line confidence (good, moderate or poor), select a colour for the line and write any notes about the line feature.

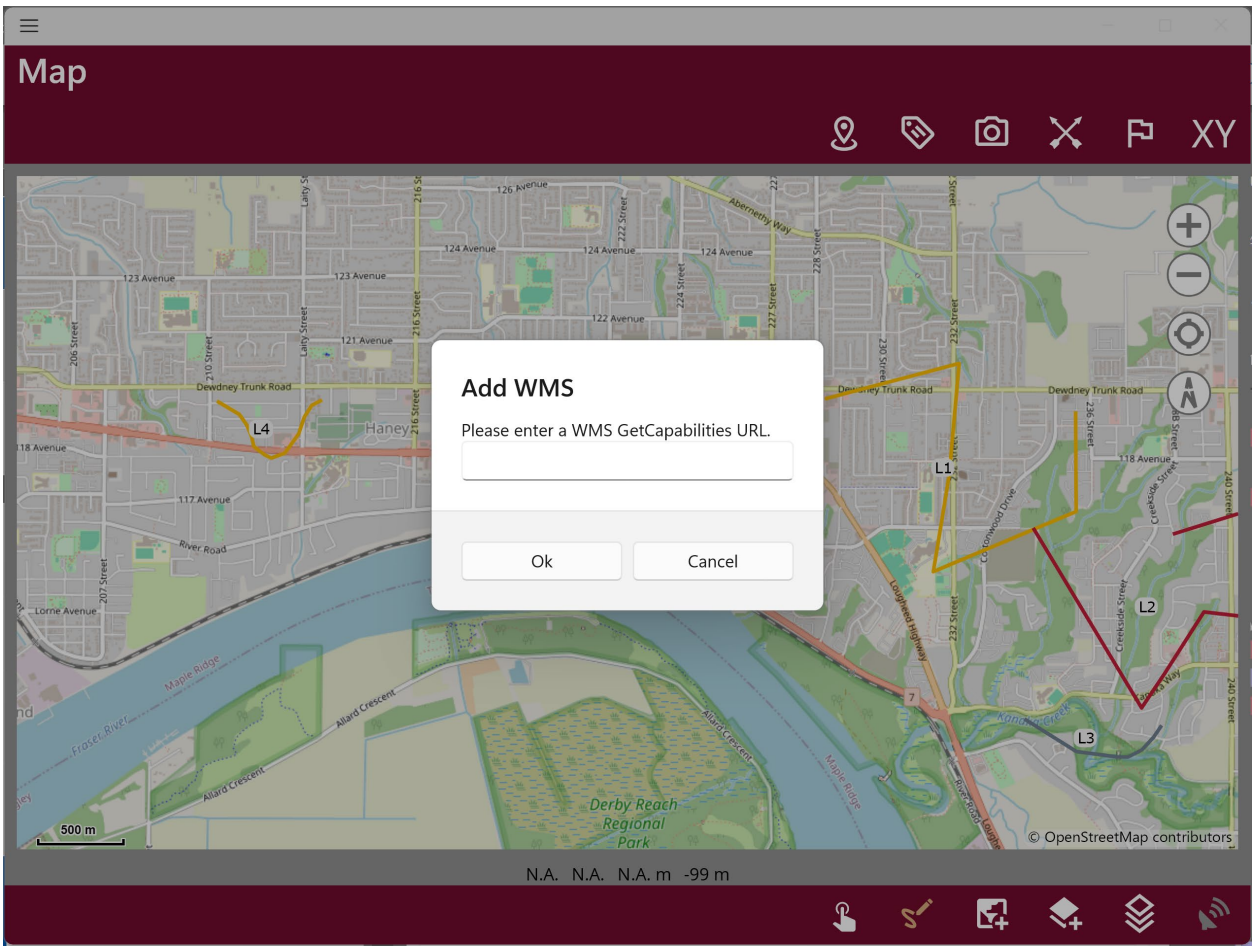

Linework
25FFF0004
Linework
Type
contact
Confidence
moderate
Line Color
amber
Notes
Bear scat, massive foot prints
Delete
Save
Copy

USING A WEB MAPPING SERVER (WMS) / UTILISATION D'UN SERVEUR DE CARTOGRAPHIE WEB (WMS)

NOTE: the use of this feature requires an Internet connection.

Tap the 'Tap to add an URL of an online Web Mapping Server (WMS).

Enter the WMS URL



The following is an excellent source for a lot of basic geology data: https://canada3d-geo.azurewebsites.net/ows?service=wms&version=1.3.0&request=GetCapabilities

USING LAYERS (TPKs, MBTILEs, GEOPACKAGES) / UTILISATION DE LAYERS (TPKs, MBTILEs, GEOPACKAGES)

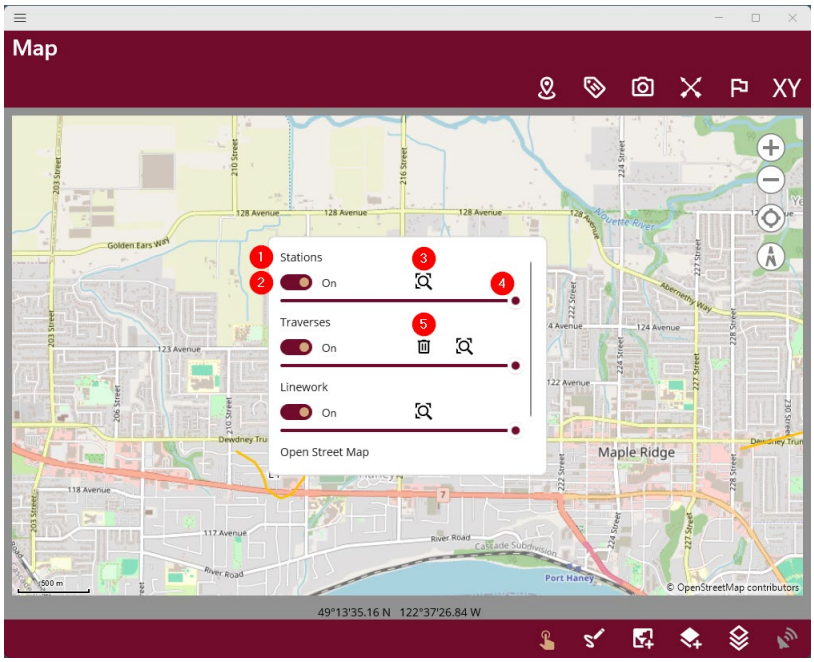
Tap on the 'Tap to add a new layer' button

Use the browser to navigate to the location of your layer and select it. Click on the OK button to add the layer.

NOTE: Loading vector features that hold more than a thousand records might affect the general performance of the app on the device, possibly causing crashes, depending on the device capabilities.

USING THE LAYER MANAGER / UTILISER LE GESTIONNAIRE DE COUCHES

Tap on 'Tap to see a list of all loaded layers and more options' to enter the Layer Manager.



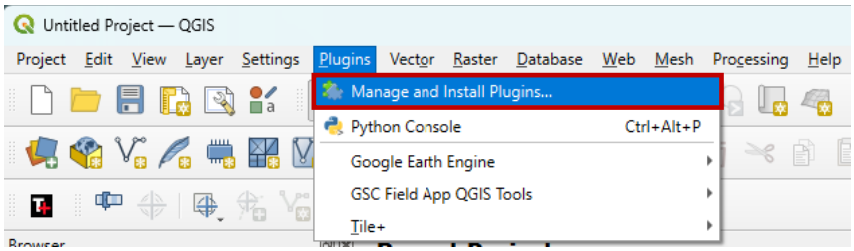
1. Name of the layer
2. Toggle the layer on/off
3. Zoom into the records in the layer
4. Transparency of the layer (drag the marker to the left to increase transparency)
5. Delete this layer and all of the data stored in it

USING THE QGIS TOOLS / UTILISATION DES OUTILS QGIS

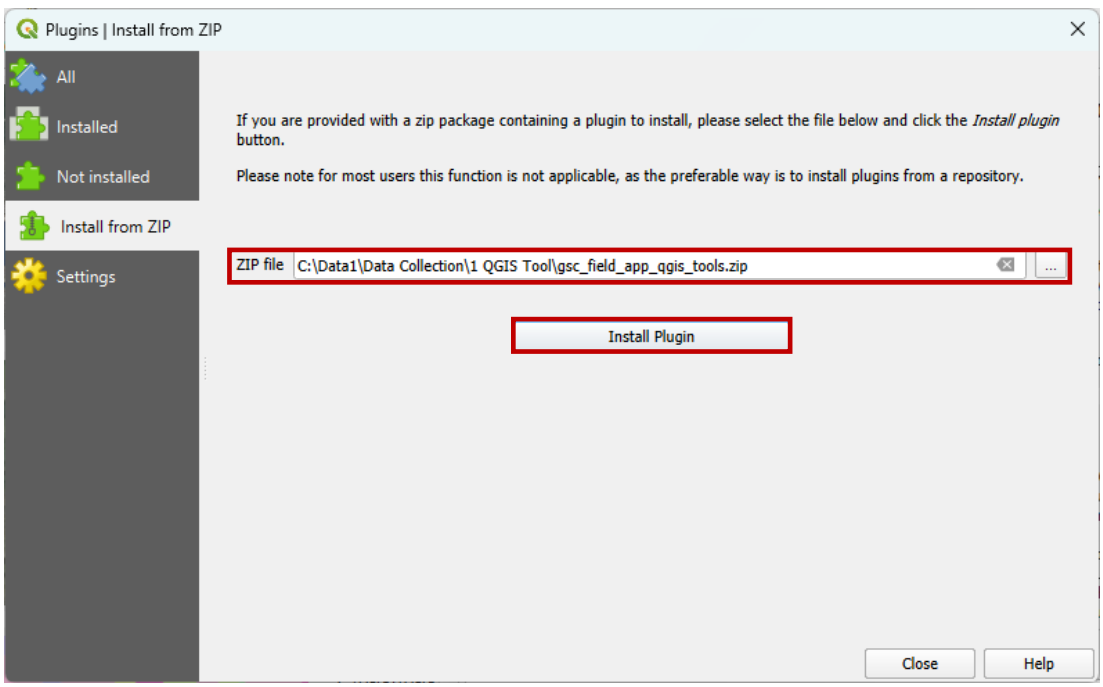
NOTE: Please contact Gabriel Huot-Vézina (gabriel.huot-vezina@rncan-nrcan.gc.ca) for access to the QGIS Tools.

Install the QGIS tools

1. Start QGIS (preferably a stable LTR release such as 3.40.4)
2. Click on **Plugins > Manage and Install Plugins…**

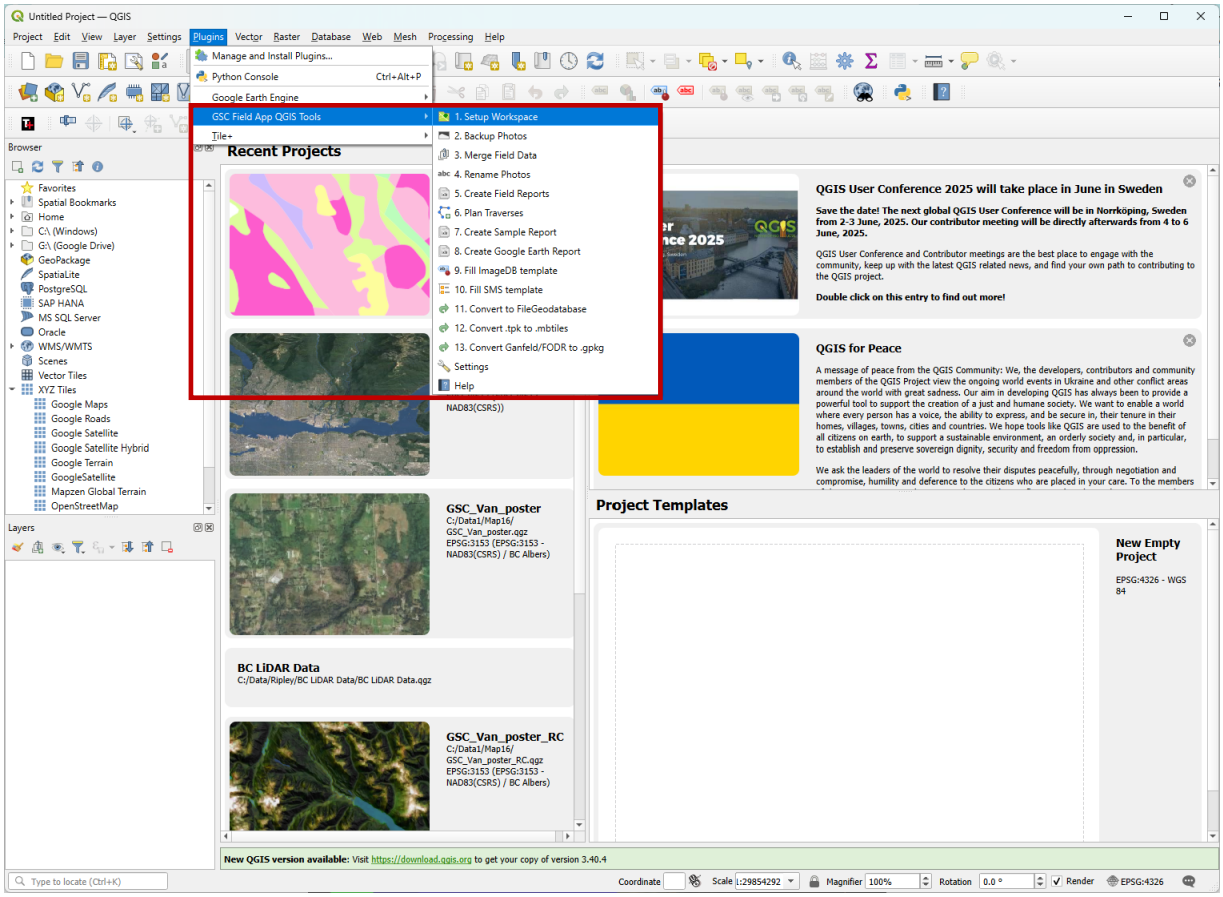


3. Direct the Plugin installer to the location of the QGIS Tools ZIP file.



4. Click on the **Install Plugin** button and click on **Yes** to accept the warning

5. Once the tools are installed, they should appear under Plugins.



6. If the QGIS Tools do not appear under Plugins, please restart QGIS.

NOTE: If your PC has more than one installation of QGIS (3.34.0 and 3.40.4 for example), the tools will install in both.